RESEARCH REPORT

Allocating AI Translation Compute for Marginal Educational Access

A global language, interlanguage, accessibility, and open-curriculum portfolio model

A global portfolio model for language, shared-core, accessibility, and open-curriculum decisions

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Abstract

AI can reduce the production cost of multilingual educational resources, but a speaker-count sort does not identify where one more edition creates the most usable access. This paper defines the relevant outcome as *marginal intelligibility*: the additional comfortable, practical educational access created for an exact language/variety/script/territory/learner profile after subtracting current native- language supply, usable academic-lingua-franca access, completed local work, format barriers, and overlap with other proposed editions. It joins 477 population observations and 83 source records to a 54-profile global recall universe, 16 bridge or shared-production architectures, 38 conservative supplemental observations, 91 profile-population join rows, 20 intensive needs studies, 24 normalized needs edges, and 100 concrete commissioning profiles. Population measures remain typed: territory, mother tongue, home use, functional use, institutional estimate and learner cohort are never silently added or treated as one measured harmed-person denominator.

The resulting Top 10 is a transparent operational commissioning sequence, not a cardinal global welfare leaderboard. In the current canonical order it comprises: 1. Bangla shared-source caregiver/ECE and Indian TVET residuals (SHC-BN); 2. Telugu applied STEM and tertiary-transition package (NAT-003); 3. Odia (NAT-010); 4. Bhojpuri (India) (NAT-125); 5. Hausa boko Nigeria–Niger package with optional established Ajami layers (SHC-HAUSA); 6. Bambara (NAT-082); 7. Standard Hindi first-year and research-methods bridge (NAT-124); 8. Peru Ashaninka Latin-script edition (EXP-026); 9. Russian learned standard (GLB-027); 10. Dari (NAT-058). The current positions of the separately discussed cases are Bahasa Indonesia: position 32; Mainland Simplified Chinese: position 34; Standard Hindi: position 7; Indian Bhojpuri: position 4; Japanese: position 37; Indonesian–Malaysian Malay localization: position 31. Every Top-100 row states its stage or unresolved stage scope, first and follow-on package, prerequisite route, delivery and accessibility requirements, existing supply, uncertainty, double-count rule and operational ordering basis. Sourced need and supply evidence are distinct from the policy choices that resolve an uncertain marginal-access partial order into a useful work sequence.

The corrected analysis does not infer that interlanguages are useless for mathematics. Formal notation and conventional proof structure plausibly increase receptive access for expert, symbol-dense tasks. The present evidence cannot turn that prior into a family-population multiplier: 24 Top-100 entries have exact normalized interlanguage/shared-core relations, but all matched cross-language reach rows remain non-rankable. Additional demographic reach is unestimated, not demonstrated absent; no family population is added. Semantic-source, terminology, formula, script and QA reuse remain separate production opportunities.

> The established Indonesian program, with its completed Open Logic corpus and a registered snapshot of 27 public course roles and 13 production roles, provides the first measured production anchor. A bounded 33-root window recorded 83,638,632,771 accounting tokens, of which 81,480,422,656 were cached input, 1,906,326,611 fresh input, and 251,883,504 output. The descendant-inclusive 6,726-task closure recorded 10,253,232,856,362 cumulative tokens but includes the roots and lacks category splits. These numbers cannot be converted directly to money, FLOPs, energy or a universal token-per-page rate. They show that cumulative program accounting is not translated-prose volume. These counters do not isolate the cost of translation, critique, correction, builds or accessibility work, so no stage-specific cost dominance is inferred.

1. Research problem

The policy problem is how to allocate a finite amount of model work so that the next bounded edition produces the largest credible increase in usable education. "Forty percent lack education in a language they understand" is not itself a language list. Walter and Benson's (2012, p. 283, Table 14.2) historical calculation assigned 3,741,110,588 people to languages used in formal education and 2,300,263,716 to languages not used, 61.9% and 38.1% of its covered population. The same table counted 97 languages above ten million speakers: 52 used in education and 45 not used. Those 45 represented 1,120,220,125 people. The chapter prints size bands, not the names of all 97 languages. The figure is directly related to the language-of-education concern but does not identify modern commissions. Crawford et al.'s separate set of 96 alphabetic EGRA assessment languages is unrelated and is not substituted for it.

The World Bank's later estimate—about 37% of students in low- and middle-income countries, roughly 370 million—uses a student/instructional-language universe and asks whether teaching occurs in a learner's first or best-understood language (World Bank, 2021). It is not Walter and Benson's global language-population denominator. Neither percentage can be multiplied by every modern language count.

Functional increase in intelligibility is more concrete: a reader who currently cannot comfortably use the available source can successfully navigate the new edition, prerequisites, notation, examples, exercises, feedback and delivery format for the intended task. This can be a child's first numeracy bridge, a vocational learner's quantitative module, a disabled student's semantic-math edition, or a researcher's access to a canonical monograph. Education includes research and continuing self-education; the audience and outcome must be stated rather than erased.

2. Formal allocation framework

The controlling recovered formalism uses four separate opportunity dimensions: R (plausible newly comfortable readers), S (scarcity of usable material), A (practical accessibility) and N (non-overlap with current comfortable access). Vitality/marginalization V, prestige-domain value P, production feasibility F and dialect/standardization risk D remain separate. R5 means more than 50 million; R4 10–50 million; R3 1–10 million; R2 100,000–1 million; R1 below 100,000; R0 no material incremental cohort identified; RU unknown. These are planning categories, not measured beneficiary counts. A plausible literature-backed judgment is admissible without experimentally measured uptake. An ordinal label is not multiplied by another ordinal label as though their product were a cardinal welfare quantity.

The new package-calibration layer specifies R/S/A/N for exact current outputs and cohorts. A formal-reasoning readership prior cannot silently become an early-years audience, a national speaker total cannot become a calculus audience, and an optional future course cannot inflate a smaller first package. S is explicitly generalized from the earlier advanced-material definition to scarcity of the selected curricular or format function at its own stage. Because R already discounts existing-language, edition and modality overlap, N documents that residual rather than subtracting the same people a second time. The baseline and cautious/favorable bands are disclosed research judgments. A separate measured-only view makes the difference from observation visible; it is not the sole legitimate way to make a recommendation.

For a secondary numeric sensitivity, let P_i be a source-defined population or learner universe, L_i the probability or bounded share able to use the target written/spoken modality, A_i practical accessibility, O_i comfortable overlap with existing editions or academic lingua francas, and D_i the exact remaining content/format deficit. A within-measure-class marginal-access sensitivity is:

$$M\_i = P\_i \times L\_i \times A\_i \times (1 - O\_i) \times D\_i.$$

Here the indexed D_i denotes a content deficit, not the formalism's dialect-risk band D. Factors must be interpreted conditionally on the same learner universe; independent marginal percentages cannot identify their intersection. This numeric diagnostic does not replace the controlling ordinal opportunity framework.

M_i is not reported as an observed harmed-person count unless every factor and joint universe is directly supported. Unknown factors remain intervals; an ignorance midpoint is labelled a sensitivity, not evidence. Territory, L1, L2, home-language, institutional and learner-cohort denominators are not cardinally compared as though they measured the same thing. Sourced observations can support bounded comparisons, but unknown marginal-access factors leave a partial order with ties or incomparable profiles. The operational allocation policy produces a usable Top 10 and Top 100 from that incomplete evidence. Its declared criteria and tie-breaking rules are decision rules, not additional measurements of educational benefit. Each emitted row retains its `ordering_basis`; the current policy diagnostic is reported in Section 3.

For a bounded product with lifecycle accounting tokens `C_i`, a compatible efficiency sensitivity is `E_i = M_i / C_i`. Source tokens, editable units, requests, fresh input, cached input, output, critique passes, correction, builds, accessibility conversion and maintenance are recorded separately. Gross accounting tokens are not hardware compute or price.

For an architecture with a shared semantic core and named outputs *j*:

`C_bundle = C_core + Σ(C_locale,j + C_script,j + C_QA,j)`.

The core has no readers of its own. Direct access is credited only to exact named outputs after overlap controls. For constructed or naturally intercomprehensible surfaces, comprehension is a function of direction, script, prior study, expertise, formal density and task. Mathematics raises the prior for expert receptive reading; it does not prove novice explanation, word-problem, writing or oral-instruction reach.

3. Data and reproducibility

The independent universe contains 54 profiles and 16 architectures. Of the profiles, 47 currently have at least one source-specific population observation and 7 remain explicitly unresolved rather than disappearing. Dispositions are: forward_residual_audit=9, ranked_in_typed_view=22, shared_core_only=2, supply_rich_negative_control=8, unresolved_no_comparable_count=13. The supplemental layer has 38 rows and the join has 91 rows. All supplement and join rows are gross/context evidence and remain non-rank-ready for unique marginal access.

The 17-step rebuild starts with normalized interlanguage staging and equal-basis augmentation, then rebuilds candidates, population evidence, needs, exact forward residuals, the canonical portfolio and its companions. The current pipeline receipt is `V1_1_DATA_PIPELINE_REBUILD_20260831.json` (SHA-256 `B59154152CC35DF16B7F26D0233C4FB4129F5AB99E121676CA0CF6DCE7A546F7`). The Top-100 companion receipt is `top100_companions_v1_1_validation.json` (SHA-256 `605A3E7E8D3E516D811FC20902D787143A19E20CD73C17388403C3D7C949E638`).

The full-register selector separates an evidence partial order from an operational work queue. A 6:3:1 cycle reserves turns for high-reach, regional-depth and endangered/prestige work. This is an explicit allocation choice, not an estimated optimum. Within a lane, it uses the currently undominated actions, then stage-specific package evidence, then the least-exposed compatible population-measure subqueue; gross-size bands, target/format specificity and finally stable identifiers resolve remaining choices. It never renames a gross population as newly comfortable readership. Literature-constrained package-specific planning judgments are now admitted: the controlling formalism asks for plausible readership ranges, not experimentally measured uptake. A separately reported measured-only view retains those judgments as priors rather than presenting them as observations. Recovered OpenLogic R/S/A/N judgments apply only to exact-profile formal-reasoning packages; they are not copied onto ECE, TVET or small residual products. Missing magnitudes remain unknown, not zero. A static evidence-front number is only a diagnostic: once a candidate's remaining dominators clear, it can compete immediately. Unknown/incomparable rows therefore do not have to be exhausted before a well-evidenced candidate in a later static layer becomes eligible. Exact duplicate packages have one owner. A canonical component register separates locale, package, stage and modality before ranking; mixed bundles must be residualized and their opportunity recalibrated. Reused components are recorded separately and cannot retain another audience. Different stages and outputs are not silently deleted.

The operational policy diagnostic is `allocation_policy_v1_1_validation.json` (SHA-256 `DA7831A1C4F01D3FC9D650C4C0F820E8B38BA2412A1FEC34D05C966BD7DC1B1C`). Its reported fields are reproduced below. A PASS status concerns its declared deterministic checks; it does not establish a measured welfare optimum.

Policy / diagnostic field	Reported value
Policy identity	educational-access-dispatch-1.1.0
Master candidates / constructed actions / selected actions	214 / 198 / 197
Exact duplicate-package rows suppressed	1
Strict evidence-dominance comparisons	71
Provisional source-informed R judgments / unresolved R	41 / 157
Package-calibration memoranda	2
Declared cycle	high_reach_underserved -> high_reach_underserved -> regional_depth -> high_reach_underserved -> high_reach_underserved -> regional_depth -> high_reach_underserved -> endangered_prestige -> high_reach_underserved -> regional_depth
Within-view gross-scope thresholds (persons)	1,000,000, 10,000,000, 50,000,000
Mechanical validation	PASS

The following is a policy-sensitivity experiment, not a confidence interval. Only the lane allocation cycle changes; the source observations and opportunity dimensions remain byte-for-byte equivalent in the computation. Positions can therefore move without any new claim about a population's educational need. This makes the discretionary part of the recommendations visible.

Action	Default 6:3:1	Balanced 1:1:1	Scale emphasis 8:1:1
SHC-BN	position 1	1	1
NAT-003	position 2	4	2
NAT-010	position 3	2	9
NAT-125	position 4	7	3
SHC-HAUSA	position 5	10	4
NAT-082	position 6	5	19
NAT-124	position 7	13	5
EXP-026	position 8	3	10
GLB-027	position 9	16	6
NAT-058	position 10	8	29
NAT-121	position 32	57	24
NAT-122	position 34	59	25
NAT-123	position 37	63	27

The next comparison changes the explicitly supplied opportunity judgments while keeping the lane schedule fixed. Cautious and favorable select the disclosed ordinal endpoints; the baseline uses an explicit analyst-selected band when supplied, never an automatically invented midpoint. Range comparison preserves the full supplied envelope. Measured-only deliberately withholds provisional judgments from active comparisons, without claiming zero access. These are conditional planning scenarios, not statistical confidence intervals or probabilities of success.

Action	Base	Cautious	Favorable	Range comparison	Measured-only
SHC-BN	position 1	11	14	25	25
NAT-003	position 2	41	1	4	4
NAT-010	position 3	3	3	30	30
NAT-125	position 4	2	2	5	5
SHC-HAUSA	position 5	1	4	2	2
NAT-082	position 6	151	152	6	6
NAT-124	position 7	44	5	7	7
EXP-026	position 8	8	8	8	8
GLB-027	position 9	4	7	9	9
NAT-058	position 10	6	6	10	10
NAT-121	position 32	45	31	12	12
NAT-122	position 34	52	39	41	24
NAT-123	position 37	55	34	24	29

21 current packages have source-informed calibrations; Appendix G gives every scope and band rationale. Other exact-profile formal-reasoning priors are retained only in their compatible domain. Observation-only census rows remain in the evidence register but are not manufactured into extra translation commissions. In particular, a US home-language count does not become a population in the language's country of origin, and a supply-rich comparison row does not automatically commission another generic formal-reasoning translation.

The component ownership correction is substantive, not a language exclusion. NAT-001 owns Bangladesh grades 2–5 recovery; SHC-BN retains caregiver-mediated preschool learning and Indian Bengali secondary/TVET. NAT-124 owns the Hindi first-year/methods package; IL-HU retains only the Indian and Pakistani Urdu outputs. Their R/S/A/N judgments are recalibrated to those active components. Caregivers are delivery intermediaries, not extra children counted as beneficiaries. The Malaysian Malay FR-2 core appears once as NAT-040; IL-IDMS remains its shared-source architecture alias rather than a second commission. Distinct school catch-up, advanced modules, regional surfaces and Jawi derivatives are not silently collapsed. SHC-SWAHILI's current population/dispatch scope is DRC/Uganda only; Tanzania/Kenya are follow-on context, not active first-package audiences.

OpenStax and Open Logic are auditable, editable baselines for zero-price reuse, not universal books. Need is measured by population profile × learner stage × subject × format × current supply.

4. Educational stages and product selection

Stage	Binding access question	Typical first product	Evaluation unit
Pre-primary	Is mathematical language available in the home/familiar register?	Oral counting, comparison, shape, pattern, caregiver activities	Child/caregiver task
Primary	Can a learner repair number, operation, fraction and measurement foundations?	Diagnostic mastery cycle with worked feedback and complete answers	Mastered prerequisite
Lower secondary	Can the learner cross into proportional reasoning, algebra, geometry and data?	Catch-up and academic-register bridge	Problem set and re-entry path
Upper secondary/TVET	Does the route connect to functions, precalculus, statistics and occupation-specific numeracy?	Applied bridge, simulations and sector cases	Course or occupational competency
Undergraduate	Are coherent prerequisite chains and proof practices available?	Calculus, linear algebra, discrete math, probability and proof sequence	Course/unit
Professional/continuing	Can readers update health, engineering, finance, management and data practice?	Versioned applied modules and notebooks	Competency/use case
Postgraduate/doctoral	Can students enter advanced seminars, monographs and research methods?	Terminology, proof and literature-navigation bridge	Course, monograph or seminar
Active researcher	Can specialists read, teach and extend frontier corpora?	Title-audited canonical source and dependency graph	Reference work/research uptake

The product baseline is teacher-independent without assuming that teachers are irrelevant: a motivated reader should be able to download once, diagnose prerequisites, follow worked examples, attempt exercises, inspect complete answers and recover from errors without an account, LMS or persistent connection. Classroom use remains an additional valid route. Semantic HTML/MathML, reflowable EPUB, tagged/print PDF, captions, audio transcripts, keyboard operation, large print and low-bandwidth bundles are separate access axes, not decoration.

The content universe is wider than mathematics. OpenStax provides an auditable starting catalog for physics, chemistry, biology, statistics, business, economics, management, health and social science. Local evidence may instead prioritize financial capability, workplace numeracy, agriculture, climate/disaster resilience, public health, computing, data practice, entrepreneurship or research methods. Advanced mathematical corpora such as EGA, SGA and FGA are educational infrastructure for specialist readers, not mass foundational substitutes.

5. Results: operational commissioning order

Position: 1 · ID: SHC-BN · Intervention: Bangla shared-source caregiver/ECE and Indian TVET residuals

Profiles: bn-Beng-BD; bn-Beng-IN

Planning R/S/A/N: R3/S3/A3/N3

Population basis: Bangla, Bangladesh: 168,419,331 (interval 168,410,840–168,427,822) [mother_tongue_or_first_language; 2022 population / 2023 survey] | Bengali, India: 97,237,669 [mother_tongue_or_first_language; 2011] | Bengali, India: 96,177,835 [same_name_mother_tongue_component; 2011]

First bounded package: Produce only the residual Bangladesh 24-week caregiver/ECE component and the Indian Bengali secondary/TVET workshop-mathematics, safety, coding, finance and entrepreneurship bridge. Link the Bangladesh primary progression to NAT-001 grades 2–5 recovery; do not recommission or count that recovery component here.

Position: 2 · ID: NAT-003 · Intervention: Telugu applied STEM and tertiary-transition package

Profiles: te-Telu-IN with Andhra Pradesh and Telangana overlays

Planning R/S/A/N: R3/S3/A4/N3

Population basis: Telugu: 81,127,740 [mother_tongue_or_first_language; 2011] | Telugu: 80,912,459 [same_name_mother_tongue_component; 2011]

First bounded package: Produce applied secondary/TVET/first-year STEM, coding, financial safety and entrepreneurship with Telugu–English terminology and offline simulations.

Position: 3 · ID: NAT-010 · Intervention: Odia

Profiles: or-Orya-IN

Planning R/S/A/N: RU/SU/AU/NU

Population basis: Odia: 37,521,324 [mother_tongue_or_first_language; 2011] | Odia: 34,059,266 [same_name_mother_tongue_component; 2011]

First bounded package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Position: 4 · ID: NAT-125 · Intervention: Bhojpuri (India)

Profiles: bho-Deva-IN

Planning R/S/A/N: R3/S3/A3/N3

Population basis: Bhojpuri, India: 50,579,447 [mother_tongue_or_first_language; 2011]

First bounded package: Create a declared versioned Devanagari Bhojpuri editorial register and a teacher-independent foundational-numeracy-through-algebra sequence with diagnostics, full worked answers, phone/offline delivery, audio, and Bhojpuri-to-Hindi bridge terminology.

Position: 5 · ID: SHC-HAUSA · Intervention: Hausa boko Nigeria–Niger package with optional established Ajami layers

Profiles: ha-Latn-NG; ha-Latn-NE; optional locally established ha-Arab layers

Planning R/S/A/N: R3/S3/A2/N3

Population basis: Hausa Boko country outputs: 58,000,000 [published_or_derived_speaker_estimate; 2022] | Hausa Boko country outputs: 94,000,000 [published_or_derived_speaker_estimate; 2022]

First bounded package: Build boko structured literacy/numeracy with diagnostics and Hausa↔English in Nigeria / Hausa↔French in Niger; add print, radio, IVR and optional established Ajami parallel text/audio.

Position: 6 · ID: NAT-082 · Intervention: Bambara

Profiles: bm-Latn-ML

Planning R/S/A/N: R2..R3/S4/A2..A3/N4

Population basis: Mali Bamanankan Latin-script edition: 9,551,561 [census_mother_tongue_persons_age_3_plus; 2022]

First bounded package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Position: 7 · ID: NAT-124 · Intervention: Standard Hindi first-year and research-methods bridge

Profiles: hi-Deva-IN

Planning R/S/A/N: R3/S3/A4/N3

Population basis: Standard Hindi, India: 322,230,097 [same_name_mother_tongue_component; 2011]

First bounded package: Produce the Hindi first-year physics, chemistry, biology, statistics, computing, economics and management bridges with research design, R/Python and open-science methods. NAT-124 owns this exact Hindi component; IL-HU reuses its semantic/formula core for two separate Urdu locale outputs without recommissioning Hindi.

Position: 8 · ID: EXP-026 · Intervention: Peru Ashaninka Latin-script edition

Profiles: cni-Latn-PE

Planning R/S/A/N: RU/SU/AU/NU

Population basis: Ashaninka: 73,567 [census_language_learned_in_childhood_age3plus_persons; 2017]

First bounded package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Position: 9 · ID: GLB-027 · Intervention: Russian learned standard

Profiles: ru-Cyrl-RU; localized successor-state overlays

Planning R/S/A/N: RU/S1/A4/N2

Population basis: Russian learned standard: 134,319,233 [functional_language_ability_or_estimate; 2021-10-01] | Russian learned standard: 132,317,159 [home_or_usual_language_use; 2021-10-01]

First bounded package: Open, accessible, no-login/offline statistics/causal inference, reproducible Python/R, research integrity, open licensing/data stewardship and AI evidence evaluation with applied tracks.

Position: 10 · ID: NAT-058 · Intervention: Dari

Profiles: prs-Arab-AF

Planning R/S/A/N: RU/SU/AU/NU

Population basis: Dari: 30,083,114 [derived_2020_total_speakers_from_official_secondary_share_and_world_bank_denominator; 2020 estimate; republished in official report 2023]

First bounded package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

The canonical Top 10 above is reproduced from `TOP_10.csv`, the exact prefix of `TOP_100.csv`; it is not a separately fixed shortlist. Its ordering is the operational policy result, while each row's evidence and uncertainty must be read on their own terms. Across the broader portfolio, Bangla shares one semantic source but requires separate Bangladesh and Indian curricula. MSA is a learned written standard and must be paired with named spoken-language scaffolds. Hindi and Urdu reuse semantics while keeping Devanagari, Nastaliq, country standards and exact population cells separate. Kiswahili and Hausa require country and orthography outputs rather than umbrella reach. Filipino is not every Philippine learner's first language. Vietnamese and Telugu enter with explicit transition packages. The Indonesian and Chinese cases are at position 32 and position 34, respectively. Their prior work is credited in the forward deficit rather than used to erase either profile from opportunity analysis; these facts alone do not prove either position.

5.1 What the Top 10 actually need

Position: 1 · ID: SHC-BN

Why now: Documented Bangladesh ECE participation gaps and Indian Bengali secondary/TVET material discontinuity support two distinct localized residuals. The Bangladesh grades 2–5 recovery commission is owned by NAT-001 and is excluded from this action's reader estimate.

Follow-on: Separate Bangladesh disaster-resilience and Indian Bengali first-year university overlays; then postgraduate research methods.

Prerequisite route: Oral-language/ECE entry -> grades 2-5 literacy/numeracy recovery -> lower-secondary re-entry -> TVET/applied pathway. | Secondary mastery diagnostic -> workshop mathematics/science -> TVET or first-year bridge -> research methods.

Access design: Print, offline HTML, audio, IVR/SMS-compatible prompts and SD-card/local-server bundle. | Offline phone/desktop course graph plus print workshop cards.

Position: 2 · ID: NAT-003

Why now: A large exact L1 population and substantial school supply point to the school-to-TVET/first-year transition, not another primary textbook series.

Follow-on: Agriculture/climate, health and SME continuing education, then postgraduate research methods and reproducibility.

Prerequisite route: Secondary prerequisite repair -> TVET/applied STEM -> first-year bilingual bridge -> professional/research methods.

Access design: Offline simulations, semantic HTML/EPUB/PDF and print practical guides.

Position: 3 · ID: NAT-010

Why now: Use ISO 639-3 `ory` in research tables; keep `or` as tooling alias.

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Access design: AX-HTML;AX-PDF;AX-OFFLINE

Position: 4 · ID: NAT-125

Why now: The exact 50,579,447 mother-tongue return establishes scale but not literacy, academic comfort, current population or the number newly served; Hindi academic-reading overlap is unmeasured.

Follow-on: After an exact supply audit, add missing secondary/TVET practical science, health, finance and computing; keep tertiary/professional work as a separately evidenced residual.

Prerequisite route: Bhojpuri oral/Devanagari entry diagnostic -> foundational numeracy -> pre-algebra/algebra -> secondary/TVET bridge -> separately audited tertiary residual.

Access design: Download-once semantic HTML/MathML, EPUB, tagged PDF, print workbook and low-data audio; no login required.

Position: 5 · ID: SHC-HAUSA

Why now: Hausa has a very large L1/L2 universe and unusually strong cross-border reuse potential, but script, literacy and country-curriculum reach are not one number.

Follow-on: Secondary/TVET science, agriculture, public health, computing, mobile-money safety, microenterprise, livestock/water and solar/GSM trades; then tertiary statistics/R/Python.

Prerequisite route: Boko/Ajami placement and oral-literacy entry -> foundational mastery -> practical TVET -> statistics/programming/open science.

Access design: Print, radio, IVR, basic-phone and offline HTML/PDF.

Position: 6 · ID: NAT-082

Why now: N'Ko literacy is a separate script derivative.

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Access design: AX-HTML;AX-PDF;AX-OFFLINE

Position: 7 · ID: NAT-124

Why now: The sharper discontinuity is higher education and professional/research continuity, not absence of all Hindi school books.

Follow-on: Current health, agriculture, cyber-finance and SME continuing education, then accessible catch-up practice only where exact gaps remain.

Prerequisite route: Bilingual secondary diagnostic -> first-year disciplinary bridge -> statistics/programming -> postgraduate research methods.

Access design: Offline bilingual HTML/EPUB/PDF plus low-cost laboratory/simulation assets.

Position: 8 · ID: EXP-026

Why now: Americas balance; exact Indigenous-language cell

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Access design: AX-HTML;AX-PDF;AX-OFFLINE

Position: 9 · ID: GLB-027

Why now: The marginal need is reuse/accessibility/current methods and exact frontier gaps, not wholesale school translation.

Follow-on: Exact modern frontier review/monograph translations and semantic-accessible editions only where inventory proves absence.

Prerequisite route: Disciplinary prerequisite check -> statistics/programming/research integrity -> reproducible professional/research work -> exact frontier gaps.

Access design: Open no-login offline semantic bundle plus source and print.

Position: 10 · ID: NAT-058

Why now: Shared source and term graph can reduce compute.

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Access design: AX-HTML;AX-PDF;AX-OFFLINE

5.2 Large-profile completeness and omitted billions

After Indian Bhojpuri was added at the exact 2011 C-16 mother-tongue count of 50,579,447, every direct person-count language observation at or above 50 million in the current official master maps to the expected universe. The 528,347,193 Census “Hindi” umbrella is never added to its same-name Hindi component or Bhojpuri and other children. The remaining billion-scale defect was a modality conflation: written `zh-Hans-CN` and spoken Putonghua. The latter now has its own profile. Applying the official rounded 80.72% communication prevalence to the mainland census denominator gives a derived sensitivity of 1,139,587,786, with a rounding-only interval of 1,139,517,198–1,139,658,374. It is not an official speaker, L1, literacy or academic- comfort count and is wholly nonadditive with written Chinese and Sinitic profiles (Ministry of Education of the People's Republic of China, 2021; National Bureau of Statistics of China, 2021).

Malaysian Malay is also explicit. The 32.4 million Malaysia total is a rounded territory ceiling, not a Malay-reader count. MABBIM supports strong terminology and production reuse with Indonesian, not blanket direct comprehension. The named Indonesian–Malaysian architecture's first action is represented once by NAT-040 at position 31; it begins by measuring and executing the `ms-Latn-MY` locale delta from the completed Indonesian semantic corpus; Brunei Rumi and Jawi outputs stay separate (Department of Statistics Malaysia, 2022; Dewan Bahasa dan Pustaka, n.d.).

5.3 Existing work and exact forward residuals

Pos.: 4 · Case: Bhojpuri (India)

Credited existing supply: Official sources evidence Hindi-medium schooling and some Bhojpuri-as-subject provision, but no continuous open Bhojpuri-medium mathematics/STEM sequence has been evidenced.

Next work: Create a declared versioned Devanagari Bhojpuri editorial register and a teacher-independent foundational-numeracy-through-algebra sequence with diagnostics, full worked answers, phone/offline delivery, audio, and Bhojpuri-to-Hindi bridge terminology.

Non-overlap: Never add Bhojpuri to the parent Hindi census umbrella; award zero automatic Hindi comprehension credit; keep Nepal, Mauritius, Magahi and Maithili outputs separate.

Pos.: 7 · Case: Standard Hindi first-year and research-methods bridge

Credited existing supply: NCERT/DIKSHA school provision is extensive; DIKSHA supports offline use and many languages; Hindi had the largest audited NIMI vocational output.

Next work: Produce the Hindi first-year physics, chemistry, biology, statistics, computing, economics and management bridges with research design, R/Python and open-science methods. NAT-124 owns this exact Hindi component; IL-HU reuses its semantic/formula core for two separate Urdu locale outputs without recommissioning Hindi.

Non-overlap: NAT-124 is the sole owner of this hi-Deva-IN first-year/professional/methods package. IL-HU owns Urdu outputs only. Exclude Bhojpuri and other Hindi-umbrella returns; neither source reuse nor bilingual overlap creates another Hindi audience.

Pos.: 31 · Case: Malaysian Malay formal reasoning via Indonesian source reuse

Credited existing supply: Malaysian educational supply exists; the verified reusable source asset is Indonesian Open Logic 722/722. Exact Malaysian terminology/curriculum deltas are distinct from Indonesian readership.

Next work: Produce the registered FR-2 Formal-reasoning core, 210 source units covering FR-1 plus first-order logic, as a Malaysian Malay locale adaptation of the completed 722/722 Indonesian Open Logic corpus. NAT-040 owns this bounded shared-source action; IL-IDMS is its architecture alias, not another commission. Retain diagnostics, complete worked answers, editable source and offline HTML/MathML/PDF.

Non-overlap: One ms-Latn-MY FR-2 commission, canonically NAT-040. IL-IDMS is an exact first-action alias retained in architecture/disposition data. Do not count Indonesian beneficiaries, the architecture itself, additional Open Logic units or Jawi derivatives as this action's new audience.

Pos.: 32 · Case: Bahasa Indonesia Evidence-to-Practice and Research Core

Credited existing supply: Indonesian program, not pilot: Open Logic 722/722 and a 40-course mathematics program with public and active-production roles.

Next work: Add six nonduplicative courses: research design/open science; applied statistics and causal reasoning in R/Python; reproducible data/AI/cybersecurity; evidence appraisal; public health/One Health; climate-smart agriculture/disaster risk.

Non-overlap: Do not duplicate the completed Open Logic corpus or already covered 40-course roles; commission only exact residual or new-domain packages.

Pos.: 34 · Case: Mainland Simplified Chinese accessibility and frontier residual

Credited existing supply: Large national platform supply; local Open Logic 722/722 and Algebra and Trigonometry 94/94 complete; Calculus I 30/55 with 25 exact units remaining; Open Logic has a deterministically evidenced accessibility residual.

Next work: Finish the exact 25-unit Calculus I residual (CALC1-0031 through CALC1-0055), then Calculus II/III and Statistics; produce semantic HTML/MathML, accessible EPUB, tagged-PDF, audio/transcript and low-bandwidth offline derivatives for completed corpora.

Non-overlap: Do not claim coverage of Wu, Yue vernacular, Xiang, Hakka, Taigi or minority-language communities; do not duplicate complete Open Logic or algebra.

Pos.: 37 · Case: Japanese (Japan)

Credited existing supply: All compulsory-school pupils receive required textbooks free; approved Japanese school series are abundant and attainment is high. Residual cohorts include 84,759 public-school pupils requiring Japanese-language instruction, 353,970 non-attending compulsory-school pupils and roughly 10% of the PIAAC age-16-65 population at or below numeracy Level 1; these cohorts overlap.

Next work: Build a no-login, download-once Japanese Open Mathematics Access and Research Bridge: curriculum-aligned diagnostics and prerequisite repair, complete solutions, accessible semantic HTML/EPUB/MathML, and easy-Japanese/furigana plus multilingual terminology for non-attending pupils, adult re-entry learners and pupils requiring Japanese-language instruction.

Non-overlap: Do not duplicate generic compulsory-school algebra or treat the 123.802M territory population as Japanese readers newly served; keep formalism-assisted foreign-language reading distinct from writing/oral-instruction access.

The Indonesian program is established, not a pilot and not a wholly completed 40-course program. The registered public snapshot records Open Logic 722/722, a 1,116-page linked reader, 40 selected course roles, 27 effective published roles and 13 production roles. Its documented page universes—19,745 teaching-package pages, 20,763 selected-corpus working pages and 27,705 rendered-universe pages—are not interchangeable, and final Indonesian pagination is unknown. Its next package therefore adds six nonduplicative evidence-to-practice and research courses rather than repeating Open Logic (OpenLogic-id, 2026; Program Matematika Indonesia, 2026).

Mainland China likewise receives no duplicate Open Logic or algebra. The current exact state is Open Logic 722/722, *Algebra and Trigonometry 2e* 94/94 and *Calculus Volume 1* 30/55. The first bounded work is CALC1-0031 through CALC1-0055, followed by Calculus II/III and Statistics, semantic accessibility, a curriculum-aligned school diagnostic overlay, a 144-hour vocational quantitative bridge, adult/workplace/ financial numeracy, academic re-entry and title-audited frontier work. Six further STEM titles are candidate scope, not six verified residual books: their completion and remaining units stay unknown until the exact works and coverage are audited. Official platform abundance rules out “China lacks native-language education” as the diagnosis; it does not prove open licences, no-login offline permanence, independent-study closure or accessibility (Ministry of Education of the People's Republic of China, 2026; National Bureau of Statistics of China, 2026; W3C, 2023, 2025).

Japan's current placement is position 37 in the operational sequence; this is not a measured estimate of its relative marginal benefit. Free compulsory textbooks, multiple approved series and high PISA/TIMSS attainment make another ordinary school textbook a low-additionality comparator. Concrete overlapping residual cohorts include 84,759 pupils requiring Japanese-language instruction, 353,970 compulsory-school pupils meeting the nonattendance definition, and roughly 10% of the PIAAC age-16–65 population at or below numeracy Level 1. Upper-secondary accessible-audio coverage was reported far below elementary coverage, and a bounded MEXT audit found independent screen-reader operation inconsistent. The commission is a Japanese Open Mathematics Access and Research Bridge: open offline self-study, easy Japanese/ruby/multilingual scaffolding, semantic mathematics, adult re-entry, proof/foreign-language reading and title-audited frontier translations (Ministry of Education of Japan, 2025, 2026a, 2026b; OECD, 2024; Sawaki, 2017).

6. Interlanguage and shared-core result

The overlap companion finds exact normalized relations for 24 of 100 portfolio entries. No matched matrix row is currently rankable for cross-language demographic reach, so the cross-language credit is conservatively zero. This is not an estimate of zero mathematical utility. It means that family ceilings, short cloze tests, names and scripts are not multiplied into reader populations without directed task evidence.

The production recommendation remains positive where reuse is coherent: Bangla, Hindi–Urdu, Arabic, Indonesian–Malaysian Malay, BCMS, Nguni, Sotho–Tswana, Persian– Dari–Tajik, Punjabi, Scandinavian, Turkic-branch and other packages can share semantic source, terminology, formulas, accessibility structure, rendering tests and QA while publishing named outputs. Interslavic remains a useful supplementary mathematics- reading and terminology experiment. Its observed short professional-text result does not establish sustained mathematical comprehension or blanket Slavic coverage.

The next research design should cross expertise (novice, undergraduate, specialist), formal density (expository prose, worked example, theorem-proof, notation-heavy reference), direction, script, prior study and task (gist, exact proposition, error-detection, problem solution). This tests the user's mathematically plausible hypothesis directly instead of importing a general-prose prior.

7. Compute evidence and planning

Boundary	Gross input	Cached input	Fresh input	Output	Total	Interpretation
33 registered owner roots	83,386,749,267	81,480,422,656	1,906,326,611	251,883,504	83,638,632,771	Exact final cumulative counters; reasoning is already inside output
6,726-task descendant-inclusive closure	not split	not split	not split	not split	10,253,232,856,362	Includes the 33 roots; never add to the root subtotal

The 33-root total partitions as cached + fresh + cache-write gross input, with cache-write zero. Reasoning output is already contained in output. The closure total is an inclusive superset, not a descendant add-on. A tail scan found 7,970 distinct cumulative-token progressions, an evidenced lower bound rather than a complete request count. These boundaries are the reason “a million gross tokens” cannot be converted directly into a million fresh tokens or into weekly monetary spend.

For a fixed 722-unit Open Logic source, the pre-existing planning scenarios are:

Scenario	Source alpha tokens	Editable units	Fresh input	Cached input	Output	Gross tokens	Status
low	367220	722	1199000	1023000	457000	2679000	Disciplined first-pass translation with targeted review; hypothetical workflow coefficients; not weekly Codex-plan consumption.
base	367220	722	4738000	5854000	1074000	11666000	Every unit receives one independent model critique and about 35 percent receives a rewritten slice; hypothetical workflow coefficients; not observed local usage.
high	367220	722	19435000	30324000	3440000	53200000	Two full critique passes substantial rewriting verbose QA and 40 percent retry allowance; conservative stress scenario; not observed local usage.

Those scenarios are hypothetical workflow coefficients, not measured Indonesian usage. They demonstrate how source preparation, per-unit calls, critique, correction, build QA and retry allowance can change gross workload. Future grants should report at least: source tokens, editable units, fresh input, cached input, output, model/request boundary, formula/structure checks, build/render checks, accessibility conversion, corrections and maintenance. Package comparisons should use the same boundary.

8. Grant and implementation design

1. **Use open, editable sources.** Global reuse requires lawful translation, correction, redistribution, printing and preservation at zero learner price.
2. **Select an exact profile and exact deficit.** Record variety, script, territory, stage, subject, format, current supply and non-overlap before production.
3. **Build once, localize explicitly.** Reuse semantic cores and toolchains while retaining named locale/script outputs and zero automatic demographic reach.
4. **Make self-study a baseline.** Include diagnostics, prerequisites, worked examples, exercises, complete answers, misconception feedback and download-once delivery.
5. **Treat accessibility as content.** Semantic math, keyboard operation, captions, transcripts, reflow, large print, Braille/speech compatibility and low-bandwidth bundles are independently commissioned and checked.
6. **Maintain separate foundational and frontier lanes.** A small specialist corpus can have high research and teaching leverage without masquerading as mass reach.
7. **Instrument the lifecycle.** Persist source identities, terminology decisions, formula/structure checks, builds, renders, model critiques, corrections, token categories and public-byte receipts.
8. **Evaluate use without creating a human-dependent release gate.** Learning and comprehension evidence should inform later revisions; complete deterministic work remains publishable while uncertainty is disclosed.

9. Limitations

Population observations span incompatible measures and years. Territory ceilings are not reader counts; L1 is not literacy; home use is not academic comfort; an official language is not universal competence; and a learner cohort is not a language population. Several expected profiles remain unresolved. Top-100 positions are a decision-aid sequence, not a measured global welfare ordering.

Needs evidence is deepest for 20 high-reach profiles. Other Top-100 rows use explicit provisional local-need fits from registered curriculum portfolios and are labelled as such. A package recommendation can be reasonable before a full local supply census, but it cannot be described as a demonstrated absence. Clinical, legal, financial, agricultural and warning information requires local versioning.

AI mathematical competence improves semantic checking relative to blind machine translation, but it is not infallibility or linguistic authority. Source-faithful formula and structure checks, terminology evidence, independent model critique, compilation, rendering and versioned correction remain necessary deterministic parts of the workflow. Missing human evidence is uncertainty, not a reason to halt a complete provisional edition.

The compute audit is unusually large but not a universal cost model. Its earliest Open Logic work is outside the registered boundary, cached inputs dominate the root total, the descendant closure lacks category splits, and page universes include different source/reference roles. No cost, energy or carbon claim is derived.

10. Conclusion

An evidence-conscious educational translation program need not choose between large languages, small endangered languages, interlanguages, accessibility and advanced research. It separates their outcomes and builds a portfolio. Large exact profiles can create vast aggregate access; regional and endangered editions create depth and prestige-domain function; semantic accessibility reaches readers whom language count misses; and research-frontier translation supports advanced education and knowledge production.

The current Top 10 is the canonical sequence reproduced in Section 5, not a claim that its members are the ten empirically best uses of compute. The broader portfolio credits completed Indonesian and Chinese work, retains Japan's specific access needs, and keeps their operational positions distinct from unmeasured marginal benefit. The analysis explicitly represents Indian Bhojpuri, separates spoken Putonghua from written Chinese, and exposes Malaysian Malay as its own output. It uses interlanguage mechanisms where they save production work and gives direct readership only where evidence supports comprehension. Most importantly, it says what to build: from caregiver numeracy and mastery recovery through TVET, statistics, financial and workplace capability, undergraduate proof, professional updating, semantic access and canonical research corpora.

The next production decision can now be bounded and auditable: choose one emitted profile, take its stated first package rather than a generic textbook, measure exact source units and workflow tokens, publish open offline accessible outputs, and update the evidence after use. That is a tractable path from AI compute to additional educational access.

Model and contributor provenance

Research synthesis, analysis, deterministic data generation, drafting and document production used **OpenAI Codex gpt-5.6-sol, Ultra**. The model identity does not replace the authorities cited below. Original source authors, OpenStax/Open Logic contributors, public-program contributors and human

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Appendix A. Full ordered Top 100: population, stage, and product

Pos.: 1 · ID: SHC-BN · Intervention: Bangla shared-source caregiver/ECE and Indian TVET residuals

Profiles: bn-Beng-BD; bn-Beng-IN

Scope: Bangladesh; West Bengal and other Indian Bengali contexts

Population context: Bangla, Bangladesh: 168,419,331 (interval 168,410,840–168,427,822) [mother_tongue_or_first_language; 2022 population / 2023 survey] | Bengali, India: 97,237,669 [mother_tongue_or_first_language; 2011] | Bengali, India: 96,177,835 [same_name_mother_tongue_component; 2011]

Stages: Bangladesh caregiver/pre-primary; Indian Bengali secondary/TVET transition

First package: Produce only the residual Bangladesh 24-week caregiver/ECE component and the Indian Bengali secondary/TVET workshop-mathematics, safety, coding, finance and entrepreneurship bridge. Link the Bangladesh primary progression to NAT-001 grades 2–5 recovery; do not recommission or count that recovery component here.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|Bangladesh residents represented by the official 2023 SEDS mother-tongue estimate|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 2 · ID: NAT-003 · Intervention: Telugu applied STEM and tertiary-transition package

Profiles: te-Telu-IN with Andhra Pradesh and Telangana overlays

Scope: Andhra Pradesh and Telangana, India

Population context: Telugu: 81,127,740 [mother_tongue_or_first_language; 2011] | Telugu: 80,912,459 [same_name_mother_tongue_component; 2011]

Stages: secondary/TVET; first-year tertiary; professional continuing education

First package: Produce applied secondary/TVET/first-year STEM, coding, financial safety and entrepreneurship with Telugu–English terminology and offline simulations.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 3 · ID: NAT-010 · Intervention: Odia

Profiles: or-Orya-IN

Scope: Odisha India

Population context: Odia: 37,521,324 [mother_tongue_or_first_language; 2011] | Odia: 34,059,266 [same_name_mother_tongue_component; 2011]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 4 · ID: NAT-125 · Intervention: Bhojpuri (India)

Profiles: bho-Deva-IN

Scope: India

Population context: Bhojpuri, India: 50,579,447 [mother_tongue_or_first_language; 2011]

Stages: foundational numeracy through algebra

First package: Create a declared versioned Devanagari Bhojpuri editorial register and a teacher-independent foundational-numeracy-through-algebra sequence with diagnostics, full worked answers, phone/offline delivery, audio, and Bhojpuri-to-Hindi bridge terminology.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|regional_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 5 · ID: SHC-HAUSA · Intervention: Hausa boko Nigeria–Niger package with optional established Ajami layers

Profiles: ha-Latn-NG; ha-Latn-NE; optional locally established ha-Arab layers

Scope: Nigeria and Niger; cross-border communities

Population context: Hausa Boko country outputs: 58,000,000 [published_or_derived_speaker_estimate; 2022] | Hausa Boko country outputs: 94,000,000 [published_or_derived_speaker_estimate; 2022]

Stages: foundational literacy/numeracy; secondary/TVET; practical professional learning

First package: Build boko structured literacy/numeracy with diagnostics and Hausa↔English in Nigeria / Hausa↔French in Niger; add print, radio, IVR and optional established Ajami parallel text/audio.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 2 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|Unspecified global L1 universe|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 6 · ID: NAT-082 · Intervention: Bambara

Profiles: bm-Latn-ML

Scope: Mali

Population context: Mali Bamanankan Latin-script edition: 9,551,561 [census_mother_tongue_persons_age_3_plus; 2022]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|unspecified_source_stratum|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 7 · ID: NAT-124 · Intervention: Standard Hindi first-year and research-methods bridge

Profiles: hi-Deva-IN

Scope: India

Population context: Standard Hindi, India: 322,230,097 [same_name_mother_tongue_component; 2011]

Stages: first-year tertiary through active researcher

First package: Produce the Hindi first-year physics, chemistry, biology, statistics, computing, economics and management bridges with research design, R/Python and open-science methods. NAT-124 owns this exact Hindi component; IL-HU reuses its semantic/formula core for two separate Urdu locale outputs without recommissioning Hindi.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|same_name_mother_tongue_component_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 8 · ID: EXP-026 · Intervention: Peru Ashaninka Latin-script edition

Profiles: cni-Latn-PE

Scope: Peru

Population context: Ashaninka: 73,567 [census_language_learned_in_childhood_age3plus_persons; 2017]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 9 · ID: GLB-027 · Intervention: Russian learned standard

Profiles: ru-Cyrl-RU; localized successor-state overlays

Scope: Europe/Central Asia

Population context: Russian learned standard: 134,319,233 [functional_language_ability_or_estimate; 2021-10-01] | Russian learned standard: 132,317,159 [home_or_usual_language_use; 2021-10-01]

Stages: professional/postgraduate/research

First package: Open, accessible, no-login/offline statistics/causal inference, reproducible Python/R, research integrity, open licensing/data stewardship and AI evidence evaluation with applied tracks.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view home_or_usual_language_use|Respondents answering language-use question|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 10 · ID: NAT-058 · Intervention: Dari

Profiles: prs-Arab-AF

Scope: Afghanistan and displaced cohorts

Population context: Dari: 30,083,114 [derived_2020_total_speakers_from_official_secondary_share_and_world_bank_denominator; 2020 estimate; republished in official report 2023]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view institutional_speaker_estimate|total_language_speakers_L1_plus_L2_nonexclusive|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 11 · ID: GLB-024 · Intervention: Latin American Spanish localized core

Profiles: es-419 with country overlays

Scope: Americas

Population context: Latin American Spanish localized core: 44,867,699 (interval 44,724,507–45,010,891) [home_or_usual_language_use; 2024] | Latin American Spanish localized core: 519,115,258 [published_or_derived_speaker_estimate; 2025]

Stages: grades 3-9

First package: Teacher-independent mastery recovery in literacy, numeracy and practical science with health, household finance, digital safety and climate; diagnostics and full answers.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view home_or_usual_language_use|United States residents age 5+|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 12 · ID: GLB-025 · Intervention: Brazilian Portuguese

Profiles: pt-BR

Scope: Americas

Population context: Brazilian Portuguese: >=265,000,000 [published_or_derived_speaker_estimate; 2026] | Brazilian Portuguese: 203,080,756 [territory_all_residents_ceiling; 2022]

Stages: adult/vocational/professional/research

First package: No-login/offline, semantic, teacher-independent mastery and transition bundles using lawfully reusable Brazilian sources.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view institutional_speaker_estimate|Unspecified global speaker universe|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 13 · ID: IL-PDT · Intervention: Persian–Dari–Tajik shared-source package

Profiles: fa-Arab-IR; prs-Arab-AF; tg-Cyrl-TJ

Scope: Iran Afghanistan Tajikistan and displaced cells separate

Population context: Iranian Persian: 79,926,270 [territory_all_residents_ceiling; 2016]

Stages: stage-specific named outputs; exact first stage follows the component needs audit

First package: Reuse one semantic source for separate Iranian Persian, Afghan Dari and Tajik Cyrillic tertiary/professional paths; keep scripts, terminology and national curricula distinct.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|Iran residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 14 · ID: NAT-028 · Intervention: Vietnamese TVET–university and research-method spine

Profiles: vi-Latn-VN

Scope: Vietnam

Population context: Vietnamese: 89,000,000 [published_or_derived_speaker_estimate; 2016 publication; population estimate vintage not stated]

Stages: secondary/TVET; undergraduate; postgraduate and continuing professional

First package: Build five coherent courses: data/programming/AI/cybersecurity; climate-smart rice/aquaculture/green skills; public health/evidence appraisal; MSME accounting/digital finance; applied biology/chemistry with low-cost labs.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 2 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view institutional_speaker_estimate|Vietnam_total_Vietnamese_speakers_L1_L2_unspecified|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 15 · ID: IL-AR · Intervention: Modern Standard Arabic core with named spoken-language scaffolds

Profiles: arb-Arab core; Egyptian, Levantine, Iraqi, Gulf, Maghrebi, Yemeni, Sudanese and Hassaniya layers

Scope: Arabic-using countries, separately localized

Population context: Modern Standard Arabic plus locale scaffolds: >=400,000,000 [published_or_derived_speaker_estimate; current UNESCO page] | Egyptian Arabic scaffold: 94,798,827 [territory_all_residents_ceiling; 2017]

Stages: early literacy; emergency education; secondary/TVET; professional and research methods

First package: In parallel, build fully vowelled leveled ECE–grade 3 MSA with local-spoken narration and offline accelerated literacy/science/health/WASH packages for crisis-affected country profiles.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 3 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view territory_all_residents_ceiling|Egypt residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 16 · ID: NAT-027 · Intervention: Sinhala

Profiles: si-Sinh-LK

Scope: Sri Lanka

Population context: Sinhala: 14,948,168 [derived_cldr_territory_functional_language_users; 2026-03-17 CLDR release; component estimate year not stated]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|functional_literate_language_users_L1_or_later_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 17 · ID: GLB-044 · Intervention: Korean, Republic of Korea

Profiles: ko-Kore-KR

Scope: Eastern Asia

Population context: Korean, Republic of Korea: 51,829,136 [territory_all_residents_ceiling; 2020]

Stages: adult/professional and research continuity

First package: Thirty-six short offline life-capability modules: health, fraud/credit/SME finance, agriculture/climate, practical science, computing and AI-output verification.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view territory_all_residents_ceiling|Republic of Korea census population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 18 · ID: EXP-028 · Intervention: Peru Shipibo-Konibo Latin-script edition

Profiles: shp-Latn-PE

Scope: Peru

Population context: Shipibo-Konibo: 34,152 [census_language_learned_in_childhood_age3plus_persons; 2017]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 19 · ID: SHC-SWAHILI · Intervention: Kiswahili DRC grades 1–3 and Uganda P4 package

Profiles: swc-Latn-CD; swh-Latn-UG

Scope: Named Kiswahili-using DRC regions (grades 1–3); Uganda (P4 Kiswahili L2)

Population context: Uganda territory population: 45,905,417 [territory_all_residents_ceiling; 2024]

Stages: DRC grades 1–3 literacy/numeracy and Uganda P4 Kiswahili-L2 support

First package: Reuse/adapt the DRC ministry's listed Kiswahili grades 1–3 and literacy materials where rights permit; complete missing worked feedback, science/health, semantic/offline and regional-variety components with French bridges. Add separate Uganda P4 L2 self-study scripts and audio; do not duplicate existing schoolbooks.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 4 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view territory_all_residents_ceiling|Uganda residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 20 · ID: NAT-047 · Intervention: Kazakh

Profiles: kk-Cyrl-KZ

Scope: Kazakhstan

Population context: Kazakh: 13,380,107 [census_native_language_of_own_nationality_persons; 2021]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|direct_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 21 · ID: NAT-004 · Intervention: Indian Tamil

Profiles: ta-Taml-IN

Scope: India

Population context: Tamil, India: 69,026,881 [mother_tongue_or_first_language; 2011] | Tamil, India: 68,888,839 [same_name_mother_tongue_component; 2011]

Stages: first-year tertiary

First package: Tamil-English first-year STEM, computing and management with worked problems, offline simulations and research methods.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 3 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 22 · ID: NAT-065 · Intervention: Amharic

Profiles: am-Ethi-ET

Scope: Ethiopia

Population context: Amharic: 56,900,000 [education_cohort; 2018 estimate] | Amharic: 21,634,396 [mother_tongue_or_first_language; 2007]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|large_national_mother_tongue|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 23 · ID: NAT-035 · Intervention: Ilocano

Profiles: ilo-Latn-PH

Scope: Philippines

Population context: Iloko: 7,098,503 [CLEAR_normalized_main_language_weighted_persons_from_IPUMS_2010_census_extract; 2010-12-31]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|national_main_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 24 · ID: NAT-046 · Intervention: Uzbek

Profiles: uz-Latn-UZ

Scope: Uzbekistan

Population context: Uzbekistan Uzbek Latin-script edition: 35,700,000 (interval 35,650,000–35,749,999) [preliminary_census_mother_tongue_persons_rounded; 2026]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|unspecified_source_stratum|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 25 · ID: NAT-038 · Intervention: Javanese

Profiles: jv-Latn-ID

Scope: Indonesia

Population context: Javanese: 68,044,660 [home_or_usual_language_use; 2010]

Stages: primary_to_secondary

First package: Oral/bilingual Javanese-to-Bahasa number-sense through prealgebra/algebra scaffold with audio and deliberate terminology transfer

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view home_or_usual_language_use|large_regional_home_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 26 · ID: EXP-004 · Intervention: India Maithili Devanagari edition

Profiles: mai-Deva-IN

Scope: India

Population context: Maithili: 13,353,347 [census_same_name_mother_tongue_component_persons; 2011]

Stages: secondary through first-year tertiary

First package: Produce the registered SB-1 Secondary data bridge for this exact named output: MV-1 plus Introductory Statistics 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 27 · ID: NAT-030 · Intervention: Burmese

Profiles: my-Mymr-MM

Scope: Myanmar and displaced cohorts

Population context: Burmese: 36,817,344 [functional_language_ability_or_estimate; 2026-03-17 CLDR release; component estimate year not stated]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view functional_language_ability_or_estimate|functional_literate_language_users_L1_or_later_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 28 · ID: NAT-044 · Intervention: Fataluku

Profiles: ddg-Latn-TL

Scope: Timor-Leste

Population context: Fataluku: 41,500 [census_mother_tongue_persons; 2015]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-1 Formal-reasoning foundation for this exact named output: methods plus sets-functions-relations plus propositional; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 29 · ID: NAT-064 · Intervention: Kenyan Swahili

Profiles: sw-Latn-KE

Scope: Kenya

Population context: Standard Kiswahili with country layers: 47,564,296 [territory_all_residents_ceiling; 2019]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|Kenya residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 30 · ID: NAT-014 · Intervention: Eastern Punjabi

Profiles: pa-Guru-IN

Scope: Punjab India

Population context: Punjabi, India: 33,124,726 [mother_tongue_or_first_language; 2011] | Punjabi, India: 31,144,095 [same_name_mother_tongue_component; 2011]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 31 · ID: NAT-040 · Intervention: Malaysian Malay formal reasoning via Indonesian source reuse

Profiles: ms-Latn-MY

Scope: Malaysia

Population context: Malaysian Malay: 32,400,000 (interval 32,350,000–32,449,999) [territory_all_residents_ceiling; 2020]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core, 210 source units covering FR-1 plus first-order logic, as a Malaysian Malay locale adaptation of the completed 722/722 Indonesian Open Logic corpus. NAT-040 owns this bounded shared-source action; IL-IDMS is its architecture alias, not another commission. Retain diagnostics, complete worked answers, editable source and offline HTML/MathML/PDF.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|Malaysia residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 32 · ID: NAT-121 · Intervention: Bahasa Indonesia Evidence-to-Practice and Research Core

Profiles: id-Latn-ID

Scope: Indonesia

Population context: Bahasa Indonesia: 248,501,794 [functional_language_ability_or_estimate; 2022]

Stages: undergraduate; professional/continuing; postgraduate and active researcher

First package: Add six nonduplicative courses: research design/open science; applied statistics and causal reasoning in R/Python; reproducible data/AI/cybersecurity; evidence appraisal; public health/One Health; climate-smart agriculture/disaster risk.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 3 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view functional_language_ability_or_estimate|population_age_5_plus|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 33 · ID: NAT-049 · Intervention: Tajik

Profiles: tg-Cyrl-TJ

Scope: Tajikistan

Population context: Tajik: 10,394,100 [derived_cldr_territory_functional_language_users; 2026-03-17 CLDR release; component estimate year not stated]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|functional_literate_language_users_L1_or_later_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 34 · ID: NAT-122 · Intervention: Mainland Simplified Chinese accessibility and frontier residual

Profiles: zh-Hans-CN

Scope: Mainland China

Population context: Standard Written Chinese, Mainland Simplified: 1,411,778,724 [territory_all_residents_ceiling; 2020-11-01]

Stages: secondary through undergraduate; accessibility; professional and research frontier

First package: Finish the exact 25-unit Calculus I residual (CALC1-0031 through CALC1-0055), then Calculus II/III and Statistics; produce semantic HTML/MathML, accessible EPUB, tagged-PDF, audio/transcript and low-bandwidth offline derivatives for completed corpora.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 3 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view territory_all_residents_ceiling|national written-standard potential population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 35 · ID: NAT-006 · Intervention: Marathi

Profiles: mr-Deva-IN

Scope: Maharashtra India

Population context: Marathi: 83,026,680 [mother_tongue_or_first_language; 2011] | Marathi: 82,801,140 [same_name_mother_tongue_component; 2011]

Stages: secondary/TVET/first-year

First package: Applied science, workshop practice, computing, finance and entrepreneurship.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 3 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 36 · ID: NAT-036 · Intervention: Hiligaynon

Profiles: hil-Latn-PH

Scope: Philippines

Population context: Hiligaynon: 6,073,883 [CLEAR_normalized_main_language_weighted_persons_from_IPUMS_2010_census_extract; 2010-12-31]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|national_main_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 37 · ID: NAT-123 · Intervention: Japanese (Japan)

Profiles: ja-Jpan-JP

Scope: Japan

Population context: Japanese, Japan: 126,146,099 [territory_all_residents_ceiling; 2020-10-01]

Stages: accessible self-study, multilingual academic bridge and postgraduate/research frontier

First package: Build a no-login, download-once Japanese Open Mathematics Access and Research Bridge: curriculum-aligned diagnostics and prerequisite repair, complete solutions, accessible semantic

HTML/EPUB/MathML, and easy-Japanese/furigana plus multilingual terminology for non-attending pupils, adult re-entry learners and pupils requiring Japanese-language instruction.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 4 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view territory_all_residents_ceiling|national written-standard potential population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 38 · ID: NAT-045 · Intervention: Baikeno

Profiles: bkx-Latn-TL

Scope: Timor-Leste

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-1 Formal-reasoning foundation for this exact named output: methods plus sets-functions-relations plus propositional; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 39 · ID: NAT-063 · Intervention: Tanzanian Swahili

Profiles: sw-Latn-TZ

Scope: Tanzania

Population context: Standard Kiswahili with country layers: 61,741,120 [territory_all_residents_ceiling; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|Tanzania residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 40 · ID: EXP-005 · Intervention: Indonesia Madurese Latin-script edition

Profiles: mad-Latn-ID

Scope: Indonesia

Population context: Madurese: 7,743,533 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 41 · ID: NAT-072 · Intervention: Nigerian Pidgin

Profiles: pcm-Latn-NG

Scope: Nigeria

Population context: Nigerian Pidgin / Naijá: 80,000,000–112,000,000 [published_or_derived_speaker_estimate; 2024]

Stages: adult/community practical numeracy with oral access; written pathway separately scoped

First package: Produce Naija/Nigerian Pidgin audio and basic-phone practical numeracy modules with spoken worked examples, money/measurement/data tasks and optional aligned Latin-script transcripts; do not presume standardized written readership.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|Nigeria oral L1/L2 user universe|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 42 · ID: NAT-034 · Intervention: Cebuano

Profiles: ceb-Latn-PH

Scope: Philippines

Population context: Cebuano: 20,697,364 [CLEAR_normalized_main_language_weighted_persons_from_IPUMS_2010_census_extract; 2010-12-31]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|national_main_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 43 · ID: NAT-005 · Intervention: Sri Lankan Tamil

Profiles: ta-Taml-LK

Scope: Sri Lanka

Population context: Tamil: 3,297,390 [derived_cldr_territory_functional_language_users; 2026-03-17 CLDR release; component estimate year not stated]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|functional_literate_language_users_L1_or_later_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 44 · ID: NAT-029 · Intervention: Thai

Profiles: th-Thai-TH

Scope: Thailand

Population context: Thai: 64,080,191 [derived_union_of_census_language_strata; 2010-09-01]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|usual_home_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 45 · ID: NAT-039 · Intervention: Sundanese

Profiles: su-Latn-ID

Scope: Indonesia

Population context: Sundanese: 32,412,752 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 46 · ID: NAT-037 · Intervention: Waray

Profiles: war-Latn-PH

Scope: Philippines

Population context: Waray: 2,694,135 [CLEAR_normalized_main_language_weighted_persons_from_IPUMS_2010_census_extract; 2010-12-31]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-1 Formal-reasoning foundation for this exact named output: methods plus sets-functions-relations plus propositional; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view other_source_defined_person_count|national_main_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 47 · ID: NAT-015 · Intervention: Western Punjabi

Profiles: pnb-Arab-PK

Scope: Pakistan

Population context: Punjabi: 88,915,544 [census_mother_tongue_persons; 2023]

Stages: secondary_to_undergraduate

First package: Shahmukhi Punjabi-to-Urdu/English algebra, functions, trigonometry, precalculus and proof bridge

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|regional_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 48 · ID: NAT-102 · Intervention: Inuinnaqtun Roman profile

Profiles: ikt-Latn-CA

Scope: Canada exact region pending

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-1 Formal-reasoning foundation for this exact named output: methods plus sets-functions-relations plus propositional; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 49 · ID: NAT-056 · Intervention: Kurmanji Kurdish

Profiles: kmr-Latn-TR

Scope: Turkey Syria and separately modeled diaspora

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 50 · ID: EXP-006 · Intervention: Indonesia Minangkabau Latin-script edition

Profiles: min-Latn-ID

Scope: Indonesia

Population context: Minangkabau: 4,232,226 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 51 · ID: NAT-001 · Intervention: Bangladesh Bangla grades 2–5 mastery-recovery package

Profiles: bn-Beng-BD

Scope: Bangladesh

Population context: Bangla, Bangladesh: 168,419,331 (interval 168,410,840–168,427,822) [mother_tongue_or_first_language; 2022 population / 2023 survey]

Stages: primary grades 2–5 mastery recovery

First package: Produce the Bangladesh Bangla 24-week grades 2–5 mastery-recovery component: number sense, operations, place value, fractions, measurement and patterns, with placement diagnostics, complete worked answers, audio and offline practice. NAT-001 is its sole commission owner; SHC-BN links to this component instead of translating it again.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 2 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view mother_tongue_or_first_language|Bangladesh residents represented by the official 2023 SEDS mother-tongue estimate|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 52 · ID: NAT-033 · Intervention: Filipino–English STEM–Kabuhayan package

Profiles: fil-Latn-PH with replaceable home-language audio/glossary tracks

Scope: Philippines

Population context: Filipino, Philippines: 109,035,343 [territory_all_residents_ceiling; 2020-05-01]

Stages: grades 7–12; Alternative Learning System; TVET; first-year tertiary

First package: Produce 120 Filipino–English micro-lessons in health, personal finance/SME management, agriculture/climate/disaster, core science, computing/data/AI and cybersecurity for grades 7–12, ALS and TESDA.

Evidence: literature_backed_high_reach_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 4 is diagnostic only (planning inference is not measurement); package-evidence policy tier stage_specific; least-dispatched compatible typed view territory_all_residents_ceiling|Philippines residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 53 · ID: NAT-032 · Intervention: Lao

Profiles: lo-Laoo-LA

Scope: Laos

Population context: Lao: 5,487,956 [derived_cldr_territory_functional_language_users; 2026-03-17 CLDR release; component estimate year not stated]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|functional_literate_language_users_L1_or_later_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 54 · ID: NAT-018 · Intervention: Sindhi

Profiles: sd-Arab-PK

Scope: Pakistan

Population context: Sindhi, Pakistan: 34,401,564 [mother_tongue_or_first_language; 2023]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|regional_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 55 · ID: NAT-069 · Intervention: Hausa

Profiles: ha-Latn-NG

Scope: Nigeria and separately tagged cross-border cells

Population context: unresolved—no comparable source-defined population count

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 56 · ID: EXP-007 · Intervention: Indonesia Banjar Latin-script edition

Profiles: bjn-Latn-ID

Scope: Indonesia

Population context: Banjar: 3,651,626 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 57 · ID: IL-HU · Intervention: Hindi–Urdu shared-core architecture: two Urdu residual outputs

Profiles: ur-Aran-IN; ur-Aran-PK

Scope: Indian and Pakistani Urdu cohorts; Hindi is a separately owned source-reuse component

Population context: Urdu, Pakistan: 22,249,307 [mother_tongue_or_first_language; 2023] | Urdu, India: 50,772,631 [mother_tongue_or_first_language; 2011] | Urdu, India: 50,725,762 [same_name_mother_tongue_component; 2011]

Stages: first-year tertiary; professional/continuing; research-methods bridge

First package: Produce two separate Urdu first-year disciplinary and research-methods bridges, ur-Aran-IN and ur-Aran-PK, using the NAT-124 Hindi/source semantic, formula and exercise core. Include physics, chemistry, biology, statistics, computing, economics, management, research design and R/Python with distinct India/Pakistan examples. Hindi production and Hindi beneficiaries are excluded from this residual action.

Evidence: source_informed_provisional_urdu_stage_need

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 3 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 58 · ID: NAT-103 · Intervention: Greenlandic

Profiles: kl-Latn-GL

Scope: Greenland

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 59 · ID: NAT-019 · Intervention: Saraiki

Profiles: skr-Arab-PK

Scope: Pakistan

Population context: Lahnda-labelled recall universe: 28,849,579 [mother_tongue_or_first_language; 2023]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|regional_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 60 · ID: NAT-050 · Intervention: Mongolian

Profiles: mn-Cyrl-MN

Scope: Mongolia

Population context: Mongolian: 3,051,962 [derived_cldr_territory_functional_language_users; 2026-03-17 CLDR release; component estimate year not stated]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view territory_all_residents_ceiling|functional_literate_language_users_L1_or_later_language|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 61 · ID: NAT-070 · Intervention: Yoruba

Profiles: yo-Latn-NG

Scope: Nigeria and diaspora

Population context: unresolved—no comparable source-defined population count

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 62 · ID: NAT-002 · Intervention: Indian Bengali

Profiles: bn-Beng-IN

Scope: India

Population context: Bengali, India: 97,237,669 [mother_tongue_or_first_language; 2011] | Bengali, India: 96,177,835 [same_name_mother_tongue_component; 2011]

Stages: primary

First package: Bangla Grades 3-8 arithmetic/fractions-to-algebra catch-up with placement tests, short mastery units and full answers

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 63 · ID: EXP-008 · Intervention: Indonesia Sasak Latin-script edition

Profiles: sas-Latn-ID

Scope: Indonesia

Population context: Sasak: 2,691,127 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 64 · ID: NAT-007 · Intervention: Gujarati

Profiles: gu-Gujr-IN

Scope: Gujarat India

Population context: Gujarati: 55,492,554 [mother_tongue_or_first_language; 2011] | Gujarati: 55,036,204 [same_name_mother_tongue_component; 2011]

Stages: primary

First package: Gujarati Grades 2-6 diagnostic remediation in place value, operations, fractions and proportional reasoning

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 65 · ID: NAT-013 · Intervention: Indian Urdu

Profiles: ur-Arab-IN

Scope: India

Population context: Urdu, India: 50,772,631 [mother_tongue_or_first_language; 2011] | Urdu, India: 50,725,762 [same_name_mother_tongue_component; 2011]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 66 · ID: EXP-009 · Intervention: Indonesia Acehnese Latin-script edition

Profiles: ace-Latn-ID

Scope: Indonesia

Population context: Acehnese: 2,550,055 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 67 · ID: NAT-008 · Intervention: Kannada

Profiles: kn-Knda-IN

Scope: Karnataka India

Population context: Kannada: 43,706,512 [mother_tongue_or_first_language; 2011] | Kannada: 43,506,272 [same_name_mother_tongue_component; 2011]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 68 · ID: NAT-105 · Intervention: Maori

Profiles: mi-Latn-NZ

Scope: New Zealand

Population context: Maori: 213,849 (interval 213,847–213,851) [census_languages_spoken_multiple_response_persons; 2023]

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 2 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 69 · ID: NAT-009 · Intervention: Malayalam

Profiles: ml-Mlym-IN

Scope: Kerala India

Population context: Malayalam: 34,838,819 [mother_tongue_or_first_language; 2011] | Malayalam: 34,776,533 [same_name_mother_tongue_component; 2011]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 70 · ID: EXP-010 · Intervention: Indonesia Betawi Latin-script edition

Profiles: bew-Latn-ID

Scope: Indonesia

Population context: Betawi: 2,244,648 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 71 · ID: NAT-012 · Intervention: Pakistani Urdu

Profiles: ur-Arab-PK

Scope: Pakistan

Population context: Urdu, Pakistan: 22,249,307 [mother_tongue_or_first_language; 2023]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view mother_tongue_or_first_language|national_language_population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 72 · ID: NAT-017 · Intervention: Pakistani Pashto

Profiles: ps-Arab-PK

Scope: Pakistan

Population context: Pashto, Pakistan and Afghanistan profiles: 43,633,946 [mother_tongue_or_first_language; 2023]

Stages: target and learner-cohort resolution; curriculum stage remains provisional

First package: First resolve the exact variety, script, orthography and learner locale using the registered source/profile evidence; retain any broader census count as a ceiling only. Then scope the appropriate registered curriculum package; this is a bounded target-resolution commission, not authorization to translate into a fabricated common standard.

Evidence: profile_resolution_action_local_need_unresolved

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view mother_tongue_or_first_language|regional_language_umbrella|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 73 · ID: EXP-011 · Intervention: Indonesia Nias Latin-script edition

Profiles: nia-Latn-ID

Scope: Indonesia

Population context: Nias: 747,168 [daily_language_at_home_age5plus_persons; 2010]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 74 · ID: GLB-028 · Intervention: German, Germany

Profiles: de-Latn-DE

Scope: Europe

Population context: German, Germany: 63,433,000 [home_or_usual_language_use; 2024] | German, Germany: 77,847,000 [home_or_usual_language_use; 2024]

Stages: target/profile resolution before stage-specific production

First package: Only exact open/accessibility/frontier residuals; do not use for Germanic minority coverage.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view home_or_usual_language_use|Germany private-household population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 75 · ID: GLB-029 · Intervention: Italian, Italy

Profiles: it-Latn-IT

Scope: Europe

Population context: Italian, Italy: 47,611,000 [home_or_usual_language_use; 2024]

Stages: target/profile resolution before stage-specific production

First package: Audit only exact open/accessibility/professional/frontier residuals.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view home_or_usual_language_use|Italy resident population age 6+|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 76 · ID: EXP-013 · Intervention: Ethiopia Somali Latin-script edition

Profiles: so-Latn-ET

Scope: Ethiopia

Population context: Somali: 4,609,274 [census_mother_tongue_persons; 2007]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 77 · ID: GLB-053 · Intervention: Putonghua, Mainland spoken standard

Profiles: cmn-Hans-CN for aligned text/audio; cmn-CN for speech-only metadata

Scope: Eastern Asia

Population context: Putonghua, Mainland spoken standard: 1,139,587,786 (interval 1,139,517,198–1,139,658,374) [functional_language_ability_or_estimate; 2020 census and 2020 survey] | Putonghua, Mainland spoken standard: 1,411,778,724 [territory_all_residents_ceiling; 2020-11-01]

Stages: target/profile resolution before stage-specific production

First package: Register spoken/audio outputs separately, preserve text/audio alignment, and award no direct written-reader or Sinitic-variety reach from this profile.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view functional_language_ability_or_estimate|Mainland residents under a cross-source communication-prevalence sensitivity|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 78 · ID: NAT-108 · Intervention: Bislama

Profiles: bi-Latn-VU

Scope: Vanuatu

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-1 Formal-reasoning foundation for this exact named output: methods plus sets-functions-relations plus propositional; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 79 · ID: GLB-026 · Intervention: Global/French regional localized standards

Profiles: fr-Latn with country overlays

Scope: Global

Population context: Global/French regional localized standards: 348,000,000 [functional_language_ability_or_estimate; 2025] | Global/French regional localized standards: 396,000,000 [functional_language_ability_or_estimate; 2025]

Stages: target/profile resolution before stage-specific production

First package: Audit country/stage-specific residuals and local-language bridges.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view functional_language_ability_or_estimate|Modeled global French-competence and schooling population|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 80 · ID: EXP-015 · Intervention: Ethiopia Sidama Latin-script edition

Profiles: sid-Latn-ET

Scope: Ethiopia

Population context: Sidama: 2,981,471 [census_mother_tongue_persons; 2007]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 81 · ID: GLB-003 · Intervention: Standard Written Chinese, Taiwan Traditional

Profiles: zh-Hant-TW

Scope: Eastern Asia

Population context: Standard Written Chinese, Taiwan Traditional: 8,801,742 (interval 8,790,849–8,812,635) [first_learned_language_derived_from_percentage; 2020] | Standard Written Chinese, Taiwan Traditional: 21,089,322 (interval 21,078,430–21,100,215) [functional_language_ability_or_estimate; 2020] | Standard Written Chinese, Taiwan Traditional: 21,786,490 [territory_age_restricted_ceiling; 2020]

Stages: target/profile resolution before stage-specific production

First package: Audit exact residual and reuse semantic source with a Taiwan locale layer.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view functional_language_ability_or_estimate|Taiwan resident nationals age 6+|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 82 · ID: GLB-034 · Intervention: Iranian Persian

Profiles: fa-Arab-IR

Scope: West/Central Asia

Population context: Iranian Persian: 79,926,270 [territory_all_residents_ceiling; 2016]

Stages: target/profile resolution before stage-specific production

First package: Resolve denominator and exact tertiary/professional/frontier residual.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view territory_all_residents_ceiling|Iran residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 83 · ID: EXP-016 · Intervention: Ethiopia Wolaytta Latin-script edition

Profiles: wal-Latn-ET

Scope: Ethiopia

Population context: Wolaytta: 1,627,955 [census_mother_tongue_persons; 2007]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 84 · ID: GLB-030 · Intervention: Polish

Profiles: pl-Latn-PL

Scope: Europe

Population context: Polish: 37,489,000 [territory_all_residents_ceiling; 2024-12-31]

Stages: target/profile resolution before stage-specific production

First package: Exact residual audit only.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view territory_all_residents_ceiling|Poland residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 85 · ID: GLB-035 · Intervention: Turkish, Turkey

Profiles: tr-Latn-TR

Scope: West Asia/Europe

Population context: Turkish, Turkey: 85,664,944 [territory_all_residents_ceiling; 2024-12-31]

Stages: target/profile resolution before stage-specific production

First package: Rank only in territory/proxy view; audit exact open/accessibility/frontier residual.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view territory_all_residents_ceiling|Residents of Türkiye|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 86 · ID: EXP-017 · Intervention: Ethiopia Hadiyya Latin-script edition

Profiles: hdy-Latn-ET

Scope: Ethiopia

Population context: Hadiyya: 1,253,894 [census_mother_tongue_persons; 2007]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 87 · ID: NAT-066 · Intervention: Oromo

Profiles: om-Latn-ET

Scope: Ethiopia and neighboring cells

Population context: Oromo exact variety profiles: 24,930,424 [mother_tongue_or_first_language; 2007]

Stages: target and learner-cohort resolution; curriculum stage remains provisional

First package: First resolve the exact variety, script, orthography and learner locale using the registered source/profile evidence; retain any broader census count as a ceiling only. Then scope the appropriate registered curriculum package; this is a bounded target-resolution commission, not authorization to translate into a fabricated common standard.

Evidence: profile_resolution_action_local_need_unresolved

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view mother_tongue_or_first_language|large_national_mother_tongue|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 88 · ID: NAT-110 · Intervention: Basque

Profiles: eu-Latn-ES

Scope: Basque Country exact territory cells

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 2 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 89 · ID: GLB-001 · Intervention: Global learned English access baseline

Profiles: en-Latn; locale outputs required

Scope: Global

Population context: Global learned English access baseline: 292,827,195 [functional_language_ability_or_estimate; 2024] | Global learned English access baseline: 53,667,911 [functional_language_ability_or_estimate; 2021] | Global learned English access baseline: 247,695,110 [home_or_usual_language_use; 2024]

Stages: target/profile resolution before stage-specific production

First package: Audit exact stage/subject/format residuals and keep English overlap out of unique-language reach.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view home_or_usual_language_use|United States residents age 5+|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 90 · ID: EXP-019 · Intervention: South Africa Sepedi Latin-script edition

Profiles: nso-Latn-ZA

Scope: South Africa

Population context: Sepedi: 5,972,255 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 91 · ID: GLB-032 · Intervention: Egyptian Arabic scaffold

Profiles: arz-Arab-EG; Saidi and other profiles separate

Scope: North Africa

Population context: Egyptian Arabic scaffold: 94,798,827 [territory_all_residents_ceiling; 2017]

Stages: target/profile resolution before stage-specific production

First package: Specify oral/audio/plain-language functions and exact orthography before reach credit.

Evidence: profile_resolution_or_exact_residual_action_from_expected_universe

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view territory_all_residents_ceiling|Egypt residents|persons; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 92 · ID: NAT-071 · Intervention: Igbo

Profiles: ig-Latn-NG

Scope: Nigeria and diaspora

Population context: unresolved—no comparable source-defined population count

Stages: target and learner-cohort resolution; curriculum stage remains provisional

First package: First resolve the exact variety, script, orthography and learner locale using the registered source/profile evidence; retain any broader census count as a ceiling only. Then scope the appropriate registered curriculum package; this is a bounded target-resolution commission, not authorization to translate into a fabricated common standard.

Evidence: profile_resolution_action_local_need_unresolved

Operational ordering basis: Declared lane cycle high_reach_underserved; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier profile_or_residual_resolution; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 93 · ID: EXP-020 · Intervention: South Africa Setswana Latin-script edition

Profiles: tn-Latn-ZA

Scope: South Africa

Population context: Setswana: 4,972,787 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 94 · ID: EXP-021 · Intervention: South Africa Sesotho Latin-script edition

Profiles: st-Latn-ZA

Scope: South Africa

Population context: Sesotho: 4,678,964 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 95 · ID: NAT-112 · Intervention: Low German

Profiles: nds-Latn-DE

Scope: Germany and separately diaspora

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-1 Formal-reasoning foundation for this exact named output: methods plus sets-functions-relations plus propositional; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 96 · ID: EXP-022 · Intervention: South Africa Xitsonga Latin-script edition

Profiles: ts-Latn-ZA

Scope: South Africa

Population context: Xitsonga: 2,784,279 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 97 · ID: EXP-023 · Intervention: South Africa siSwati Latin-script edition

Profiles: ss-Latn-ZA

Scope: South Africa

Population context: siSwati: 1,692,719 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 98 · ID: EXP-024 · Intervention: South Africa Tshivenda Latin-script edition

Profiles: ve-Latn-ZA

Scope: South Africa

Population context: Tshivenda: 1,480,565 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 99 · ID: NAT-113 · Intervention: Faroese

Profiles: fo-Latn-FO

Scope: Faroe Islands

Population context: unresolved—no comparable source-defined population count

Stages: upper-secondary through undergraduate formal reasoning

First package: Produce the registered FR-2 Formal-reasoning core for this exact named output: FR-1 plus first-order-logic; adaptation D2; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle endangered_prestige; no remaining dominance predecessors in the planning view; static opportunity layer 2 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Pos.: 100 · ID: EXP-025 · Intervention: South Africa isiNdebele Latin-script edition

Profiles: nr-Latn-ZA

Scope: South Africa

Population context: isiNdebele: 1,044,377 [census_language_most_often_spoken_in_household_persons_age_1_plus; 2022]

Stages: secondary through first-year tertiary

First package: Produce the registered MV-1 Minimum viable quantitative curriculum for this exact named output: Algebra and Trigonometry 2e; adaptation D3; include placement diagnostics, complete worked answers, editable source, and download-once delivery.

Evidence: registered_curriculum_package_provisional_local_need_fit

Operational ordering basis: Declared lane cycle regional_depth; no remaining dominance predecessors in the planning view; static opportunity layer 1 is diagnostic only (planning inference is not measurement); package-evidence policy tier registered_provisional; least-dispatched compatible typed view unknown|unknown|unknown; within-view gross stratum, target/route specificity, then stable ID. Gross scope is not newly comfortable reach.

Appendix B. Full ordered Top 100: continuation, prerequisites, and access

Pos.: 1 · Intervention: Bangla shared-source caregiver/ECE and Indian TVET residuals

Follow-on: Separate Bangladesh disaster-resilience and Indian Bengali first-year university overlays; then postgraduate research methods.

Prerequisite route: Oral-language/ECE entry -> grades 2-5 literacy/numeracy recovery -> lower-secondary re-entry -> TVET/applied pathway. | Secondary mastery diagnostic -> workshop mathematics/science -> TVET or first-year bridge -> research methods.

Delivery: Print, offline HTML, audio, IVR/SMS-compatible prompts and SD-card/local-server bundle. | Offline phone/desktop course graph plus print workshop cards.

Teacher-independent requirements: Daily diagnostic cycle, full worked answers, caregiver scripts and catch-up placement/re-entry map. | Worked cases, complete answers, simulated labs and Bengali-English terminology.

Accessibility: Unicode Bangla, semantic math, TTS testing, captions, large-print and low-data audio. | Semantic Bangla/MathML, captions, audio, high-contrast print and keyboard navigation.

Non-overlap: SHC-BN owns caregiver/ECE and Indian Bengali secondary/TVET only. NAT-001 owns Bangladesh grades 2–5 recovery. NAT-002 Indian Bengali grades 3–8 catch-up is a distinct stage-specific action. Do not sum learner/caregiver roles or ordinal R bands into a measured total.

Pos.: 2 · Intervention: Telugu applied STEM and tertiary-transition package

Follow-on: Agriculture/climate, health and SME continuing education, then postgraduate research methods and reproducibility.

Prerequisite route: Secondary prerequisite repair -> TVET/applied STEM -> first-year bilingual bridge -> professional/research methods.

Delivery: Offline simulations, semantic HTML/EPUB/PDF and print practical guides.

Teacher-independent requirements: Placement, full solutions, terminology crosswalk, low-cost laboratories and career cases.

Accessibility: Telugu screen-reader/font testing, semantic math, captions and reflow.

Non-overlap: Do not average Andhra Pradesh and Telangana outcomes or treat one state curriculum as the other.

Pos.: 3 · Intervention: Odia

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Below 50M on verified measures.

Pos.: 4 · Intervention: Bhojpuri (India)

Follow-on: After an exact supply audit, add missing secondary/TVET practical science, health, finance and computing; keep tertiary/professional work as a separately evidenced residual.

Prerequisite route: Bhojpuri oral/Devanagari entry diagnostic -> foundational numeracy -> pre-algebra/algebra -> secondary/TVET bridge -> separately audited tertiary residual.

Delivery: Download-once semantic HTML/MathML, EPUB, tagged PDF, print workbook and low-data audio; no login required.

Teacher-independent requirements: Entry diagnostic; Bhojpuri oral-to-Devanagari support; complete worked feedback; explicit Hindi bridge glossary; reversible Bihar/eastern-UP terminology overlays.

Accessibility: Devanagari font/TTS/screen-reader tests, semantic mathematics, reflow, captions/transcripts, large print and keyboard navigation.

Non-overlap: Never add Bhojpuri to the parent Hindi census umbrella; award zero automatic Hindi comprehension credit; keep Nepal, Mauritius, Magahi and Maithili outputs separate.

Pos.: 5 · Intervention: Hausa boko Nigeria–Niger package with optional established Ajami layers

Follow-on: Secondary/TVET science, agriculture, public health, computing, mobile-money safety, microenterprise, livestock/water and solar/GSM trades; then tertiary statistics/R/Python.

Prerequisite route: Boko/Ajami placement and oral-literacy entry -> foundational mastery -> practical TVET -> statistics/programming/open science.

Delivery: Print, radio, IVR, basic-phone and offline HTML/PDF.

Teacher-independent requirements: Structured phonics/mastery, full feedback, country terminology and practical projects.

Accessibility: Audio, large print, semantic boko, optional local Ajami and captions/transcripts.

Non-overlap: Never invent one universal Ajami or convert oral user estimates into written educational reach.

Pos.: 6 · Intervention: Bambara

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Manding package saves compute; named Bambara output claims only Bambara cells.

Pos.: 7 · Intervention: Standard Hindi first-year and research-methods bridge

Follow-on: Current health, agriculture, cyber-finance and SME continuing education, then accessible catch-up practice only where exact gaps remain.

Prerequisite route: Bilingual secondary diagnostic -> first-year disciplinary bridge -> statistics/programming -> postgraduate research methods.

Delivery: Offline bilingual HTML/EPUB/PDF plus low-cost laboratory/simulation assets.

Teacher-independent requirements: Complete exercises/answers, terminology crosswalk, prerequisites, research datasets and reproducible notebooks.

Accessibility: Devanagari screen-reader testing, semantic math, captions, reflow and print.

Non-overlap: NAT-124 is the sole owner of this hi-Deva-IN first-year/professional/methods package. IL-HU owns Urdu outputs only. Exclude Bhojpuri and other Hindi-umbrella returns; neither source reuse nor bilingual overlap creates another Hindi audience.

Pos.: 8 · Intervention: Peru Ashaninka Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate childhood-language category; do not add to aggregate Quechua/Aimara rows or to other measures without a common-universe proof.

Pos.: 9 · Intervention: Russian learned standard

Follow-on: Exact modern frontier review/monograph translations and semantic-accessible editions only where inventory proves absence.

Prerequisite route: Disciplinary prerequisite check -> statistics/programming/research integrity -> reproducible professional/research work -> exact frontier gaps.

Delivery: Open no-login offline semantic bundle plus source and print.

Teacher-independent requirements: Exercises/answers, reproducible notebooks, datasets and bilingual search/navigation where relevant.

Accessibility: Cyrillic semantic text/math, captions, screen-reader testing and reflow.

Non-overlap: Russian competence is not blanket coverage for Indigenous, displaced, diaspora or successor-state communities.

Pos.: 10 · Intervention: Dari

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Iranian Persian yields zero credit until directional register comprehension is measured.

Pos.: 11 · Intervention: Latin American Spanish localized core

Follow-on: No-login/offline/accessibility conversion of high-value OER, then exact nursing, computing, business, research-method and frontier gaps.

Prerequisite route: Grades 3-9 diagnostic recovery -> country secondary transition -> professional or first-year path -> exact frontier residual.

Delivery: Country-localized offline HTML/EPUB/PDF and print.

Teacher-independent requirements: Diagnostics, misconception repair, full answers and country examples/regulation.

Accessibility: Semantic text/math, captions, TTS, reflow and low-bandwidth assets.

Non-overlap: `es-ES` does not dictate Latin American curricula or professional terminology; global Spanish totals are nonadditive models.

Pos.: 12 · Intervention: Brazilian Portuguese

Follow-on: Research methods/reproducibility and exact advanced-content gaps.

Prerequisite route: Adult/secondary placement -> teacher-independent transition bundle -> professional/research methods -> exact advanced residual.

Delivery: No-login offline bundle with editable source and print.

Teacher-independent requirements: Complete answers, diagnostic progression and current professional examples.

Accessibility: Semantic text/math, captions, screen-reader support and reflow.

Non-overlap: Brazil is separate from Portugal, Angola, Mozambique and other PALOP outputs.

Pos.: 13 · Intervention: Persian–Dari–Tajik shared-source package

Follow-on: Resolve exact counts and measure script/terminology delta

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: No one script/register or additive comfort count

Pos.: 14 · Intervention: Vietnamese TVET–university and research-method spine

Follow-on: Quarterly frontier-to-field briefs plus postgraduate research design, causal inference, systematic review and reproducibility.

Prerequisite route: Secondary quantitative/digital diagnostic -> TVET/university applied courses -> research design and reproducibility.

Delivery: Offline interoperable HTML/EPUB/PDF plus low-cost labs/notebooks.

Teacher-independent requirements: Prerequisite map, complete solutions, applied cases and bilingual search terminology.

Accessibility: Semantic math, captions, screen-reader testing and low-bandwidth download.

Non-overlap: Vietnamese material does not replace exact oral/home-language overlays for minoritized early-years learners.

Pos.: 15 · Intervention: Modern Standard Arabic core with named spoken-language scaffolds

Follow-on: Country-specific disciplinary literacy, TVET and research-method paths in health, climate/agriculture, finance, computing, statistics and reproducibility.

Prerequisite route: Local spoken-language oral scaffold -> vowelled MSA literacy -> disciplinary MSA -> TVET/tertiary -> research methods.

Delivery: Offline print/radio/SD/local-server bundle plus maintained professional pathways.

Teacher-independent requirements: Diagnostics, complete feedback, local spoken narration, country curriculum/re-entry maps and versioning.

Accessibility: Unicode RTL/bidi metadata, logical reading order, TTS-tested vowelization, captions, sign-language video where relevant and semantic math.

Non-overlap: Never award blanket MSA comprehension or collapse Arabic varieties and non-Arabic home-language communities.

Pos.: 16 · Intervention: Sinhala

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Own-language cell only; English comfort is not assumed from official bilingualism.

Pos.: 17 · Intervention: Korean, Republic of Korea

Follow-on: Statistics, R/Python, open licensing, data management, research integrity and applied-domain laboratories.

Prerequisite route: Short placement and digital-safety diagnostics -> adult capability modules -> statistics/programming/research methods.

Delivery: Download-once offline bundle with print/simple-language cards.

Teacher-independent requirements: Short mastery checks, full explanations, locally current examples and maintenance dates.

Accessibility: Semantic Korean text, captions, audio, large type, high contrast and key Korean Sign Language safety/health videos.

Non-overlap: No inference about DPRK follows from ROK supply; do not pool the territories.

Pos.: 18 · Intervention: Peru Shipibo-Konibo Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate childhood-language category; do not add to aggregate Quechua/Aimara rows or to other measures without a common-universe proof.

Pos.: 19 · Intervention: Kiswahili DRC grades 1–3 and Uganda P4 package

Follow-on: Follow-on context only: separately scope Tanzania swh-Latn-TZ and Kenya swh-Latn-KE secondary/advanced outputs; their populations do not enter this first action. Extend separately evidenced country routes into Kiswahili↔English/French biology, chemistry, climate/agriculture, health, computing and finance, then TVET and statistics/R/Python/open-science paths.

Prerequisite route: DRC grades 1–3 literacy/numeracy entry -> exact regional-variety and French-bridge support -> missing worked feedback; Uganda P4 Kiswahili-L2 entry -> self-study scripts and offline audio.

Delivery: Print/audio/offline HTML with country curriculum overlay.

Teacher-independent requirements: Teacher and self-study scripts, diagnostics, worked answers, French/English bridges and locally relevant cases.

Accessibility: Captions, audio, semantic text/math, large print and offline delivery.

Non-overlap: Do not use East African Standard Kiswahili as blanket coverage for DRC varieties or non-Kiswahili L1 communities. Tanzania/Kenya outputs are follow-on context only and add no current first-package audience.

Pos.: 20 · Intervention: Kazakh

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not mix current Cyrillic users with future Latin projection.

Pos.: 21 · Intervention: Indian Tamil

Follow-on: Health, climate/agriculture, cyber-finance and SME professional modules, then accessible school practice where needed.

Prerequisite route: Secondary diagnostic -> Tamil-English first-year bridge -> reproducible methods -> postgraduate/frontier navigation.

Delivery: Offline semantic course graph plus simulations and print.

Teacher-independent requirements: Full solutions, bilingual terminology, research datasets and reproducible notebooks.

Accessibility: Tamil font/TTS testing, semantic math, captions and reflow.

Non-overlap: Sri Lankan Tamil remains a separate profile and curriculum.

Pos.: 22 · Intervention: Amharic

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Broad estimate is not an exact written-reader count.

Pos.: 23 · Intervention: Ilocano

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Count exact language-speaking household evidence only in its original measure.

Pos.: 24 · Intervention: Uzbek

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Latin and Cyrillic literacy must be separate modifiers; Turkish gives zero coverage credit.

Pos.: 25 · Intervention: Javanese

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Explicit subset/overlap edge with Indonesian is mandatory.

Pos.: 26 · Intervention: India Maithili Devanagari edition

Follow-on: After the bounded SB-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> MV-1 -> SB-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Nested in parent POP-IN-010; never sum parent and component. Nepal Maithili is a distinct territory observation and profile.

Pos.: 27 · Intervention: Burmese

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Territory includes non-Burmese L1 communities.

Pos.: 28 · Intervention: Fataluku

Follow-on: After the bounded FR-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> FR-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Own-community cell only; no Tetum coverage deduction without evidence.

Pos.: 29 · Intervention: Kenyan Swahili

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Global headline is not a deduplicated written-reader count; DRC Congo Swahili is distinct.

Pos.: 30 · Intervention: Eastern Punjabi

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: No cross-script population credit.

Pos.: 31 · Intervention: Malaysian Malay formal reasoning via Indonesian source reuse

Follow-on: After the FR-2 component, scope the remaining non-FR-2 Open Logic modules and separate Malaysian Jawi derivatives individually; neither is included in this first action's reader estimate.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: One ms-Latn-MY FR-2 commission, canonically NAT-040. IL-IDMS is an exact first-action alias retained in architecture/disposition data. Do not count Indonesian beneficiaries, the architecture itself, additional Open Logic units or Jawi derivatives as this action's new audience.

Pos.: 32 · Intervention: Bahasa Indonesia Evidence-to-Practice and Research Core

Follow-on: Maintained professional updates for MSME finance/management, health CPD, renewable energy/water, agriculture/aquaculture and quarterly frontier briefs.

Prerequisite route: Use completed Indonesian Open Logic and the relevant available exact-course prerequisites -> statistics/programming entry diagnostic and prerequisite repair -> evidence-to-practice core -> maintained professional updates; completion of the whole 40-course program is not required.

Delivery: Download-once offline HTML/PWA/local server, EPUB, tagged PDF and source package.

Teacher-independent requirements: Complete exercises/answers, diagnostics, executable notebooks, maintenance cadence and evidence-update dates.

Accessibility: Semantic MathML, screen-reader navigation, captions/audio, low-bandwidth mode and print.

Non-overlap: Do not duplicate the completed Open Logic corpus or already covered 40-course roles; commission only exact residual or new-domain packages.

Pos.: 33 · Intervention: Tajik

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Iranian Persian gives zero Tajik credit without script-adjusted comprehension evidence.

Pos.: 34 · Intervention: Mainland Simplified Chinese accessibility and frontier residual

Follow-on: Add a four-band school diagnostic overlay, a 144-hour vocational quantitative bridge, adult/workplace/financial numeracy and academic re-entry, then title-audited frontier mathematics.

Prerequisite route: Exact completed-corpus inventory -> Calculus I module prerequisites -> professional/research pathway prerequisites.

Delivery: No-account, download-once semantic HTML/MathML, EPUB, tagged PDF, audio/transcripts, chapter bundles, print assets and offline source archive.

Teacher-independent requirements: Diagnostics, complete worked answers, executable notebooks where relevant, explicit prerequisites, mastery/re-entry routing and maintenance versioning.

Accessibility: True Unicode; tested CJK fonts and ToUnicode; semantic MathML; keyboard and screen-reader navigation; captions/transcripts; reflow, large print, math-speech and Braille compatibility.

Non-overlap: Do not claim coverage of Wu, Yue vernacular, Xiang, Hakka, Taigi or minority-language communities; do not duplicate complete Open Logic or algebra.

Pos.: 35 · Intervention: Marathi

Follow-on: Climate-resilient agriculture, health/SME professional modules and research methods.

Prerequisite route: Grade-level diagnostic -> secondary recovery -> TVET/first-year bridge -> professional/research methods.

Delivery: Offline bilingual course graph and printable practice/lab pack.

Teacher-independent requirements: Diagnostic prerequisite repair, worked solutions and Marathi-English terminology.

Accessibility: Devanagari semantic text/math, captions, reflow and large print.

Non-overlap: Keep foundational rural assessment evidence separate from all Marathi speakers.

Pos.: 36 · Intervention: Hiligaynon

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Keep distinct from Cebuano and generic Bisaya labels.

Pos.: 37 · Intervention: Japanese (Japan)

Follow-on: Add a university proof/foreign-language-reading bridge and translate only title-audited missing or non-open EGA/SGA/FGA and other frontier works; do not duplicate ordinary compulsory-school mathematics.

Prerequisite route: Grades 5-9 diagnostic repair -> upper-secondary proof language -> undergraduate algebra -> commutative algebra -> algebraic geometry/frontier corpus.

Delivery: No-login offline HTML5/EPUB 3/MathML, tagged PDF, low-bandwidth ZIP, print, Japanese mathematical speech and tactile/embossable diagram assets.

Teacher-independent requirements: Placement checks, worked feedback, full solutions, prerequisite graph, mastery checkpoints and Japanese-English-French research-navigation terminology.

Accessibility: Furigana/plain-Japanese modes, semantic math, Japanese speech strings, Braille-compatible math, tactile diagrams, dyslexia/low-vision settings, captions, keyboard navigation and reflow.

Non-overlap: Do not duplicate generic compulsory-school algebra or treat the 123.802M territory population as Japanese readers newly served; keep formalism-assisted foreign-language reading distinct from writing/oral-instruction access.

Pos.: 38 · Intervention: Baikeno

Follow-on: After the bounded FR-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> FR-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Own-community cell only.

Pos.: 39 · Intervention: Tanzanian Swahili

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Global headline is not a deduplicated written-reader count; DRC Congo Swahili is distinct.

Pos.: 40 · Intervention: Indonesia Madurese Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 41 · Intervention: Nigerian Pidgin

Follow-on: Extend into locally evidenced livelihood, financial-safety and health-data modules; use explicit orthography and reading-demand profiles before a text-heavy algebra course.

Prerequisite route: Oral number and measurement diagnostic -> short spoken worked examples -> practice with answers -> optional transcript/text bridge.

Delivery: Downloadable audio, radio/IVR scripts, basic-phone prompts and low-bandwidth transcript/PDF companions.

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Nigeria territory and oral users do not establish standardized written readership.

Pos.: 42 · Intervention: Cebuano

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not merge Cebuano and Bisaya household labels without source-defined mapping.

Pos.: 43 · Intervention: Sri Lankan Tamil

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Keep separate from ta-Taml-IN and avoid double-counting diaspora residents.

Pos.: 44 · Intervention: Thai

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Derived union assumptions and minority-language overlap must remain visible.

Pos.: 45 · Intervention: Sundanese

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Indonesian overlap and politeness-level design are explicit uncertainty terms.

Pos.: 46 · Intervention: Waray

Follow-on: After the bounded FR-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> FR-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Own-language cell only.

Pos.: 47 · Intervention: Western Punjabi

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not equate every Punjabi return with one Western Punjabi variety or standardized readership.

Pos.: 48 · Intervention: Inuinnaqtun Roman profile

Follow-on: After the bounded FR-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> FR-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Keep distinct from syllabic Inuktitut and other Inuit varieties.

Pos.: 49 · Intervention: Kurmanji Kurdish

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not merge with Sorani Zazaki or Turkish; diaspora cells must be residence-deduplicated.

Pos.: 50 · Intervention: Indonesia Minangkabau Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 51 · Intervention: Bangladesh Bangla grades 2–5 mastery-recovery package

Follow-on: Link to SHC-BN caregiver/ECE and separately scoped secondary progression; do not duplicate its Indian Bengali secondary/TVET package.

Prerequisite route: Short oral/numeracy placement -> learner-specific missing foundations -> grades 2–5 worked practice and mastery checks; no teacher or online service required.

Delivery: Print, offline HTML, audio, IVR/SMS-compatible prompts and SD-card/local-server bundle.

Teacher-independent requirements: Daily diagnostic cycle, full worked answers, caregiver scripts and catch-up placement/re-entry map.

Accessibility: Unicode Bangla, semantic math, TTS testing, captions, large-print and low-data audio.

Non-overlap: One owner for bn-Beng-BD grades 2–5 recovery: NAT-001. SHC-BN caregiver/ECE and Indian Bengali TVET are distinct components, not another copy of this work. Population ceilings and attainment deficits are not counted as measured causal benefits.

Pos.: 52 · Intervention: Filipino–English STEM–Kabuhayan package

Follow-on: Open science, biostatistics, public health, programming/data, accounting/management and climate-smart agriculture/renewable-energy spine.

Prerequisite route: Grades 7-12/ALS diagnostic -> STEM-Kabuhayan modules -> TVET/first-year bridge -> professional/research spine.

Delivery: Offline local-server/USB/SD package plus print and low-data audio.

Teacher-independent requirements: Full answers, practical projects, terminology, mastery checks and local livelihood cases.

Accessibility: Captions, semantic text/math, screen-reader support and replaceable home-language audio/glossaries.

Non-overlap: Filipino is not every learner's first language; do not count replaceable home-language tracks as one Filipino audience.

Pos.: 53 · Intervention: Lao

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: No Thai cross-language credit without directional comprehension evidence.

Pos.: 54 · Intervention: Sindhi

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Shared RTL tooling is not intelligibility.

Pos.: 55 · Intervention: Hausa

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: 60M and 94M source universes differ; optional Ajami has separate unknown readership.

Pos.: 56 · Intervention: Indonesia Banjar Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 57 · Intervention: Hindi–Urdu shared-core architecture: two Urdu residual outputs

Follow-on: Current health, agriculture, cyber-finance, SME and postgraduate reproducibility modules, maintaining distinct India/Pakistan examples and law.

Prerequisite route: Locale-specific Urdu–English secondary diagnostic -> first-year disciplinary bridge -> statistics/programming -> research-methods exercises; reuse source content, not presumed Hindi reading ability.

Delivery: Separate Indian/Pakistani Nastaliq offline HTML/MathML, EPUB and PDF, with accessible logical reading order, low-cost labs and downloadable notebooks.

Teacher-independent requirements: Worked answers, diagnostic entry, Urdu–English terminology crosswalk, locally appropriate examples, reproducible notebooks and no mandatory teacher or live platform.

Accessibility: Nastaliq/RTL shaping, bidirectional formula ordering, semantic math, keyboard/reflow support, extraction and screen-reader checks; captions/transcripts and printable fallback.

Non-overlap: IL-HU owns two Urdu locale outputs only. NAT-124 owns the Hindi component. HRN-004 is Hindi need evidence, not independently established Urdu need. Urdu India/Pakistan ceilings are separate and nonadditive with functional-language umbrellas; the shared source has no additional audience.

Pos.: 58 · Intervention: Greenlandic

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: No Inuktut bridge credit and no Danish comfort assumption.

Pos.: 59 · Intervention: Saraiki

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Overlaps Punjabi, Saraiki, Hindko and classification conventions.

Pos.: 60 · Intervention: Mongolian

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Keep Cyrillic Mongolia distinct from traditional-script communities.

Pos.: 61 · Intervention: Yoruba

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Nigeria territory and unjoined literacy percentages cannot be multiplied without matching strata.

Pos.: 62 · Intervention: Indian Bengali

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not use one locale-neutral Bangla edition.

Pos.: 63 · Intervention: Indonesia Sasak Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 64 · Intervention: Gujarati

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Functional total is not additive to L1.

Pos.: 65 · Intervention: Indian Urdu

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not merge with Pakistan Urdu or Standard Hindi readership.

Pos.: 66 · Intervention: Indonesia Acehnese Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 67 · Intervention: Kannada

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: C-17 L2/L3 total cannot be compared as L1.

Pos.: 68 · Intervention: Maori

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Deduct English comfort by cohort; no Polynesian-family reach.

Pos.: 69 · Intervention: Malayalam

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Do not inflate using Kerala territory without language evidence.

Pos.: 70 · Intervention: Indonesia Betawi Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 71 · Intervention: Pakistani Urdu

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: National status is not universal comfortable Nastaliq readership.

Pos.: 72 · Intervention: Pakistani Pashto

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Resolve exact production profile -> check source-stated learner prerequisites -> scope a bounded content package.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mixed denominator and dialect/orthography profiles cannot form one rank.

Pos.: 73 · Intervention: Indonesia Nias Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate named BPS daily-home-language cell; the master table does not assert portfolio additivity, so do not sum it with grouped Indonesian rows or infer cross-language coverage.

Pos.: 74 · Intervention: German, Germany

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Home use is not a literacy test and does not cover all Germanic minorities.

Pos.: 75 · Intervention: Italian, Italy

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: No direct Italian literacy census in located source.

Pos.: 76 · Intervention: Ethiopia Somali Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate 2007 mother-tongue cell; do not add cross-border observations or family-language proxies without explicit disjoint-universe accounting.

Pos.: 77 · Intervention: Putonghua, Mainland spoken standard

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: The 1,139,587,786 sensitivity is 80.72% of the mainland census denominator, not an official speaker count, L1 count, home-language count, literacy measure or academic-comfort count. It is wholly nonadditive with zh-Hans-CN and all Sinitic profiles.

Pos.: 78 · Intervention: Bislama

Follow-on: After the bounded FR-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> FR-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: No Tok Pisin or English automatic coverage.

Pos.: 79 · Intervention: Global/French regional localized standards

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Modeled Francophone totals include heterogeneous competence and schooling.

Pos.: 80 · Intervention: Ethiopia Sidama Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate 2007 mother-tongue cell; do not add cross-border observations or family-language proxies without explicit disjoint-universe accounting.

Pos.: 81 · Intervention: Standard Written Chinese, Taiwan Traditional

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Taiwan Guoyu and traditional-character literacy are not mainland Hans reach.

Pos.: 82 · Intervention: Iranian Persian

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Do not merge Dari and Tajik populations or scripts.

Pos.: 83 · Intervention: Ethiopia Wolaytta Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate 2007 mother-tongue cell; do not add cross-border observations or family-language proxies without explicit disjoint-universe accounting.

Pos.: 84 · Intervention: Polish

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Rounded survey shares are not exact population cells.

Pos.: 85 · Intervention: Turkish, Turkey

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Territory and language-neutral literacy are not Turkish reader counts.

Pos.: 86 · Intervention: Ethiopia Hadiyya Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Separate 2007 mother-tongue cell; do not add cross-border observations or family-language proxies without explicit disjoint-universe accounting.

Pos.: 87 · Intervention: Oromo

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Resolve exact production profile -> check source-stated learner prerequisites -> scope a bounded content package.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Oromia territory and Oromo macrolanguage are not one edition population.

Pos.: 88 · Intervention: Basque

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Territory and dominant-language overlaps are explicit deductions.

Pos.: 89 · Intervention: Global learned English access baseline

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Global L2/use estimates are not academic-reading comfort and overlap nearly every other profile.

Pos.: 90 · Intervention: South Africa Sepedi Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Pos.: 91 · Intervention: Egyptian Arabic scaffold

Follow-on: After target/profile resolution, select a bounded content package from the registered source inventory.

Prerequisite route: Resolve the exact target, population and existing supply before assigning a curriculum prerequisite chain.

Delivery: research dossier and source inventory only until the target profile is resolved

Teacher-independent requirements: For any later content edition: diagnostics, complete worked answers, self-check feedback and no mandatory teacher or live service.

Accessibility: For any later edition: semantic structure and mathematics, keyboard/reflow support, tested fonts/text extraction, captions/transcripts and printable low-bandwidth fallback.

Non-overlap: Egypt territory is not Egyptian-Arabic reader population; regional varieties remain.

Pos.: 92 · Intervention: Igbo

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Resolve exact production profile -> check source-stated learner prerequisites -> scope a bounded content package.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Continuum and standardized literacy require evidence.

Pos.: 93 · Intervention: South Africa Setswana Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Pos.: 94 · Intervention: South Africa Sesotho Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Pos.: 95 · Intervention: Low German

Follow-on: After the bounded FR-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> FR-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: No pan-Germanic or Dutch coverage credit.

Pos.: 96 · Intervention: South Africa Xitsonga Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Pos.: 97 · Intervention: South Africa siSwati Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Pos.: 98 · Intervention: South Africa Tshivenda Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Pos.: 99 · Intervention: Faroese

Follow-on: After the bounded FR-2 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry -> FR-1 -> FR-2.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Excluded from mainland Scandinavian bridge credit.

Pos.: 100 · Intervention: South Africa isiNdebele Latin-script edition

Follow-on: After the bounded MV-1 package, extend only after an itemized local supply and stage audit; preserve the same named profile and non-overlap rules.

Prerequisite route: Diagnostic entry and any source-stated prerequisites -> MV-1.

Delivery: AX-HTML;AX-PDF;AX-OFFLINE

Teacher-independent requirements: Placement checks, explicit prerequisite repair, complete worked examples and answers, self-check feedback, and no required teacher or live platform.

Accessibility: Semantic headings and mathematics/MathML, keyboard navigation, captions/transcripts for media, reflow, tested fonts and text extraction, low-bandwidth assets, and printable fallback.

Non-overlap: Mutually exclusive within the coded Table 3.8 household-language denominator; additive only inside that table, never with another South Africa language measure.

Appendix C. Interlanguage/shared-core crosswalk and exact forward work

Pos.: 1 · ID: SHC-BN

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High same-script semantic, terminology, formula, accessibility and QA reuse; two separately localized curricula and examples. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 2 · ID: NAT-003

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Reuse source semantics and labs across two state overlays; each curriculum crosswalk remains distinct. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 3 · ID: NAT-010

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 4 · ID: NAT-125

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 5 · ID: SHC-HAUSA

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic core reuse; country orthographies and every Ajami layer stay explicit and separately validated. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 6 · ID: NAT-082

Matrix rows: EDGE-MAN-001

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 7 · ID: NAT-124

Matrix rows: EDGE-HU-001

Mechanisms: dual_script_register

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic/formula reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: FWD-HI-FR2 FR-2 Formal Reasoning Core at D3: complete=210/210; residual=0 | FWD-HI-OTHER Uncovered Hindi subject, stage, and accessibility residuals: scope unresolved

Forward deficit: FWD-HI-FR2=0.000000 [exact_completed_corpus] | FWD-HI-OTHER=unmeasured [unknown_until_exact_target_work_format_is_registered]

Next action: FWD-HI-FR2: Do not retranslate FR-2; commission only a separately demonstrated subject, stage, or format residual. | FWD-HI-OTHER: Name and audit an exact missing work, stage, and format before computing a residual score.

Pos.: 8 · ID: EXP-026

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 9 · ID: GLB-027

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 10 · ID: NAT-058

Matrix rows: EDGE-PDT-002

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic reuse with script/locale deltas Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 11 · ID: GLB-024

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 12 · ID: GLB-025

Matrix rows: EDGE-ROM-002

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 13 · ID: IL-PDT

Matrix rows: EDGE-PDT-001;EDGE-PDT-002;EDGE-PDT-003

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic reuse with script/locale deltas Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 14 · ID: NAT-028

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single locale output with strong reuse of open labs, code and data assets across subjects. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 15 · ID: IL-AR

Matrix rows: EDGE-AR-001;EDGE-AR-002;EDGE-AR-003;EDGE-AR-004;EDGE-AR-005;EDGE-AR-006;EDGE-AR-007;EDGE-AR-008

Mechanisms: pluricentric_shared_core;accessibility_derivative

Credited cross-language reach (not estimated utility): 0

Compute reuse: High formal-register, RTL, terminology and source reuse; every spoken scaffold remains a named output. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 16 · ID: NAT-027

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 17 · ID: GLB-044

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 18 · ID: EXP-028

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 19 · ID: SHC-SWAHILI

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High standard-core reuse, but every country/DRC regional surface is separately rendered and audited. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 20 · ID: NAT-047

Matrix rows: EDGE-TUR-005

Mechanisms: natural_intercomprehension_reuse

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 21 · ID: NAT-004

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Shared formula, layout and QA only unless measured Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 22 · ID: NAT-065

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 23 · ID: NAT-035

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 24 · ID: NAT-046

Matrix rows: EDGE-TUR-009

Mechanisms: natural_intercomprehension_reuse

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 25 · ID: NAT-038

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 26 · ID: EXP-004

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 27 · ID: NAT-030

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 28 · ID: NAT-044

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 29 · ID: NAT-064

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High standard-core reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 30 · ID: NAT-014

Matrix rows: EDGE-PNJ-001

Mechanisms: dual_script_register

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic and transliteration assistance Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 31 · ID: NAT-040

Matrix rows: EDGE-IDMS-002

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: IL-IDMS Indonesian–Malaysian Malay production architecture: reuse the completed Indonesian semantic/formula source for the 210-unit FR-2 Malaysian output. Locale-delta token savings are unmeasured. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 32 · ID: NAT-121

Matrix rows: EDGE-IDMS-001

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Reuse the established terminology, build, QA and offline infrastructure; Malaysian Malay localization is a separate later output. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: FWD-ID-OLP Open Logic complete target set: complete=722/722; residual=0 | FWD-ID-PROGRAM13 Nonpublic roles in the registered 40-course program: complete=27/40; residual=13 | FWD-ID-ACCESS Accessibility, offline-delivery, and maintenance residuals: scope unresolved

Forward deficit: FWD-ID-OLP=0.000000 [exact_completed_corpus] | FWD-ID-PROGRAM13=unmeasured [unknown_heterogeneous_role_weights] | FWD-ID-ACCESS=unmeasured [unknown_until_exact_format_gap_is_registered]

Next action: FWD-ID-OLP: Do not retranslate; preserve and maintain the completed corpus. | FWD-ID-PROGRAM13: Select an exact role, work, version, and format; then measure its remaining source tokens. | FWD-ID-ACCESS: Audit each published role for semantic HTML/MathML, screen-reader, offline, and maintenance gaps.

Pos.: 33 · ID: NAT-049

Matrix rows: EDGE-PDT-003

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic reuse with script/locale deltas Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 34 · ID: NAT-122

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Very high standardized written Chinese/toolchain reuse; Taiwan/Hong Kong locale outputs remain separate. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: FWD-CN-OLP Open Logic complete target set: complete=722/722; residual=0 | FWD-CN-AT2E Algebra and Trigonometry 2e: complete=94/94; residual=0 | FWD-CN-CALC1 Calculus Volume 1: complete=30/55; residual=25 | FWD-CN-STEM6 Six further fixed STEM books: complete=unknown/6; residual=unknown | FWD-CN-ACCESS-FRONTIER Semantic accessibility and demonstrated advanced/research-frontier gaps: scope unresolved

Forward deficit: FWD-CN-OLP=0.000000 [exact_completed_corpus] | FWD-CN-AT2E=0.000000 [exact_completed_corpus] | FWD-CN-CALC1=unmeasured [unknown_until_residual_source_tokens_are_measured] | FWD-CN-STEM6=unmeasured [unknown_work_weight] | FWD-CN-ACCESS-FRONTIER=unmeasured [unknown_until_exact_target_work_format_is_registered]

Next action: FWD-CN-OLP: Do not retranslate; commission only a separately demonstrated format or semantic-access derivative. | FWD-CN-AT2E: Do not retranslate; audit only accessibility, maintenance, or edition-version residuals. | FWD-CN-CALC1: Finish the exact residual after measuring residual source tokens; do not infer D from module count alone. | FWD-CN-STEM6: Audit exact titles, current unit coverage, source tokens, and local supply before ordering. | FWD-CN-ACCESS-FRONTIER: Define exact format or advanced corpus first; then measure its source and residual units.

Pos.: 35 · ID: NAT-006

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 36 · ID: NAT-036

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 37 · ID: NAT-123

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: FWD-JA-FR2 FR-2 Formal Reasoning Core at D3: complete=unknown/210; residual=unknown | FWD-JA-STAGE Proof bridge, semantic accessibility, postgraduate, monograph, and research-frontier gaps: scope unresolved

Forward deficit: FWD-JA-FR2=unmeasured [unknown_partial_or_missing_fraction] | FWD-JA-STAGE=unmeasured [unknown_until_exact_target_work_format_is_registered]

Next action: FWD-JA-FR2: Audit exact local/open equivalents and any partial FR-2 units before deciding whether FR-2 itself is the next product. | FWD-JA-STAGE: Commission only the exact stage/corpus/format gap established by the needs audit; do not infer a generic school-algebra need.

Pos.: 38 · ID: NAT-045

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 39 · ID: NAT-063

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High standard-core reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 40 · ID: EXP-005

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 41 · ID: NAT-072

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 42 · ID: NAT-034

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 43 · ID: NAT-005

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Shared formula, layout and QA only unless measured Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 44 · ID: NAT-029

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 45 · ID: NAT-039

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 46 · ID: NAT-037

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 47 · ID: NAT-015

Matrix rows: EDGE-PNJ-002

Mechanisms: dual_script_register

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic and transliteration assistance Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 48 · ID: NAT-102

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 49 · ID: NAT-056

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 50 · ID: EXP-006

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 51 · ID: NAT-001

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High same-script semantic reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 52 · ID: NAT-033

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Reuse semantic sources and STARBOOKS-style offline packaging; home-language tracks are separate outputs. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 53 · ID: NAT-032

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 54 · ID: NAT-018

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 55 · ID: NAT-069

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Strong shared-source potential Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 56 · ID: EXP-007

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 57 · ID: IL-HU

Matrix rows: EDGE-HU-001;EDGE-HU-002;EDGE-HU-003

Mechanisms: dual_script_register

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic, formula, exercise and terminology-graph reuse; separate scripts, render QA and locale registers. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 58 · ID: NAT-103

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 59 · ID: NAT-019

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 60 · ID: NAT-050

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 61 · ID: NAT-070

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 62 · ID: NAT-002

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: High same-script semantic reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 63 · ID: EXP-008

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 64 · ID: NAT-007

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 65 · ID: NAT-013

Matrix rows: EDGE-HU-003

Mechanisms: dual_script_register

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic/formula reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 66 · ID: EXP-009

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 67 · ID: NAT-008

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Shared formula, layout and QA only unless measured Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 68 · ID: NAT-105

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 69 · ID: NAT-009

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Shared formula, layout and QA only unless measured Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 70 · ID: EXP-010

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 71 · ID: NAT-012

Matrix rows: EDGE-HU-002

Mechanisms: dual_script_register

Credited cross-language reach (not estimated utility): 0

Compute reuse: High semantic/formula reuse Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 72 · ID: NAT-017

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 73 · ID: EXP-011

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 74 · ID: GLB-028

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 75 · ID: GLB-029

Matrix rows: EDGE-ROM-004

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 76 · ID: EXP-013

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 77 · ID: GLB-053

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 78 · ID: NAT-108

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 79 · ID: GLB-026

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 80 · ID: EXP-015

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 81 · ID: GLB-003

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 82 · ID: GLB-034

Matrix rows: EDGE-PDT-001

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 83 · ID: EXP-016

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 84 · ID: GLB-030

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 85 · ID: GLB-035

Matrix rows: EDGE-TUR-001

Mechanisms: natural_intercomprehension_reuse

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 86 · ID: EXP-017

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 87 · ID: NAT-066

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 88 · ID: NAT-110

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 89 · ID: GLB-001

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 90 · ID: EXP-019

Matrix rows: EDGE-SOT-001

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 91 · ID: GLB-032

Matrix rows: EDGE-AR-002

Mechanisms: accessibility_derivative

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 92 · ID: NAT-071

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 93 · ID: EXP-020

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 94 · ID: EXP-021

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 95 · ID: NAT-112

Matrix rows: EDGE-GER-003

Mechanisms: natural_intercomprehension_reuse

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 96 · ID: EXP-022

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 97 · ID: EXP-023

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 98 · ID: EXP-024

Matrix rows: none

Mechanisms: none

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 99 · ID: NAT-113

Matrix rows: EDGE-GER-010

Mechanisms: natural_intercomprehension_reuse

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Pos.: 100 · ID: EXP-025

Matrix rows: EDGE-NGU-004

Mechanisms: pluricentric_shared_core

Credited cross-language reach (not estimated utility): 0

Compute reuse: Single named output; source, notation and accessibility tooling may be reused, but no extra reader reach is inferred. Package-specific token savings are unmeasured unless separately quantified in a named compute receipt.

Current completion: No exact work-level completion row is registered in Table C.

Forward deficit: unregistered_not_zero

Next action: Register an exact work, stage and format before assigning a forward content deficit.

Appendix D. Expected global profiles and dispositions

ID: GLB-001 · Profile: Global learned English access baseline

Tag: en-Latn; locale outputs required

Kind: learned_lingua_franca

Region: Global

Band: >=1B functional ceiling

Disposition: supply_rich_negative_control

Population warning: Global L2/use estimates are not academic-reading comfort and overlap nearly every other profile.

Next action: Audit exact stage/subject/format residuals and keep English overlap out of unique-language reach.

ID: GLB-002 · Profile: Standard Written Chinese, Mainland Simplified

Tag: zh-Hans-CN

Kind: learned_written_standard

Region: Eastern Asia

Band: >=1B territory; >=1B derived functional sensitivity

Disposition: forward_residual_audit

Population warning: Territory, Putonghua communication and simplified-character use are different universes; not spoken-Sinitic mutual intelligibility.

Next action: Finish registered partial works and commission only exact open, accessible, professional or frontier gaps.

ID: GLB-003 · Profile: Standard Written Chinese, Taiwan Traditional

Tag: zh-Hant-TW

Kind: pluricentric_written_standard

Region: Eastern Asia

Band: regional constituent

Disposition: forward_residual_audit

Population warning: Taiwan Guoyu and traditional-character literacy are not mainland Hans reach.

Next action: Audit exact residual and reuse semantic source with a Taiwan locale layer.

ID: GLB-004 · Profile: Wu varieties and written-access profiles

Tag: wuu-Hans-CN; exact variety outputs unresolved

Kind: macrolanguage_resolution

Region: Eastern Asia

Band: >=50M recall

Disposition: unresolved_no_comparable_count

Population warning: Standard Written Chinese schooling does not make spoken Wu identical to Mandarin or prove vernacular written demand.

Next action: Resolve exact varieties, modality and product need before any rank.

ID: GLB-005 · Profile: Yue/Cantonese written and spoken profiles

Tag: yue-Hant-HK; yue-Hans-CN; locale outputs

Kind: language_register_split

Region: Eastern Asia

Band: >=50M recall

Disposition: unresolved_no_comparable_count

Population warning: Formal written Chinese and written vernacular Cantonese are not the same target.

Next action: Build separate HK/Guangdong profile inventory before commissioning.

ID: GLB-006 · Profile: Xiang varieties

Tag: hsn-Hans-CN; exact standard unresolved

Kind: macrolanguage_resolution

Region: Eastern Asia

Band: 40-50M watch

Disposition: unresolved_no_comparable_count

Population warning: Do not substitute the Sinitic family or mainland population.

Next action: Resolve exact community, written modality and incremental use.

ID: GLB-007 · Profile: Standard Hindi, India

Tag: hi-Deva-IN

Kind: standard_variety

Region: South Asia

Band: >=250M exact same-name L1

Disposition: forward_residual_audit

Population warning: Bhojpuri and other named returns are nested in the census Hindi umbrella.

Next action: Commission exact bilingual tertiary, research-method, professional and accessibility residuals.

ID: GLB-008 · Profile: Urdu, Pakistan

Tag: ur-Aran-PK

Kind: national_standard

Region: South Asia

Band: >=50M functional/bridge recall

Disposition: ranked_in_typed_view

Population warning: National status is not universal comfortable Nastaliq readership.

Next action: Audit secondary-to-tertiary and professional gaps with offline Nastaliq rendering.

ID: GLB-009 · Profile: Urdu, India

Tag: ur-Aran-IN

Kind: regional_standard

Region: South Asia

Band: >=50M exact L1

Disposition: ranked_in_typed_view

Population warning: Do not merge with Pakistan Urdu or Standard Hindi readership.

Next action: Audit exact state/stage open supply and terminology transitions.

ID: GLB-010 · Profile: Bangla, Bangladesh

Tag: bn-Beng-BD

Kind: national_standard

Region: South Asia

Band: >=150M

Disposition: ranked_in_typed_view

Population warning: Derived L1 is not an exact census language count; Bangladesh and India need separate overlays.

Next action: Prioritize ECE/grades 2-5 recovery, emergency continuity, then TVET/applied pathways.

ID: GLB-011 · Profile: Bengali, India

Tag: bn-Beng-IN

Kind: regional_standard

Region: South Asia

Band: >=90M exact L1

Disposition: ranked_in_typed_view

Population warning: Do not use one locale-neutral Bangla edition.

Next action: Prioritize secondary/TVET and Bengali-English first-year bridges.

ID: GLB-012 · Profile: Telugu

Tag: te-Telu-IN

Kind: regional_standard

Region: South Asia

Band: >=80M exact L1

Disposition: ranked_in_typed_view

Population warning: Andhra Pradesh and Telangana require separate curriculum crosswalks.

Next action: Applied secondary/TVET/first-year STEM, coding, finance and entrepreneurship.

ID: GLB-013 · Profile: Marathi

Tag: mr-Deva-IN

Kind: regional_standard

Region: South Asia

Band: >=80M exact L1

Disposition: ranked_in_typed_view

Population warning: Functional total is not additive to L1.

Next action: Secondary/TVET and first-year applied science, computing, finance and agriculture.

ID: GLB-014 · Profile: Tamil, India

Tag: ta-Taml-IN

Kind: regional_standard

Region: South Asia

Band: >=50M exact L1

Disposition: ranked_in_typed_view

Population warning: Sri Lankan Tamil is a distinct locale and population cell.

Next action: Tamil-English first-year STEM, management and research methods.

ID: GLB-015 · Profile: Gujarati

Tag: gu-Gujr-IN

Kind: regional_standard

Region: South Asia

Band: >=50M exact L1

Disposition: ranked_in_typed_view

Population warning: Functional total is not additive to L1.

Next action: Audit microenterprise, applied science, finance and first-year transition.

ID: GLB-016 · Profile: Kannada

Tag: kn-Knda-IN

Kind: regional_standard

Region: South Asia

Band: >=50M functional; L1 below threshold

Disposition: ranked_in_typed_view

Population warning: C-17 L2/L3 total cannot be compared as L1.

Next action: Audit TVET/tertiary and professional residual before selecting package.

ID: GLB-017 · Profile: Malayalam

Tag: ml-Mlym-IN

Kind: regional_standard

Region: South Asia

Band: 40-50M watch

Disposition: ranked_in_typed_view

Population warning: Do not inflate using Kerala territory without language evidence.

Next action: Audit exact advanced/open/accessibility gaps.

ID: GLB-018 · Profile: Odia

Tag: ory-Orya-IN

Kind: regional_standard

Region: South Asia

Band: 40-50M watch

Disposition: ranked_in_typed_view

Population warning: Below 50M on verified measures.

Next action: Audit secondary/TVET/tertiary discontinuity.

ID: GLB-019 · Profile: Punjabi, Pakistan

Tag: pa-Arab-PK; Aran when Nastaliq asserted

Kind: census_label_to_profile_resolution

Region: South Asia

Band: >=80M exact source label

Disposition: unresolved_no_comparable_count

Population warning: Do not equate every Punjabi return with one Western Punjabi variety or standardized readership.

Next action: Resolve variety and orthography, then commission a separate Pakistan output.

ID: GLB-020 · Profile: Punjabi, India

Tag: pa-Guru-IN

Kind: regional_standard

Region: South Asia

Band: 40-50M watch

Disposition: ranked_in_typed_view

Population warning: No cross-script population credit.

Next action: Audit exact secondary/tertiary residual and synchronized dual-script production.

ID: GLB-021 · Profile: Lahnda-labelled recall universe

Tag: und-Arab-PK-Lahnda; resolve Saraiki/Hindko/Punjabi labels

Kind: invalid_aggregation

Region: South Asia

Band: >=50M recall

Disposition: unresolved_no_comparable_count

Population warning: Overlaps Punjabi, Saraiki, Hindko and classification conventions.

Next action: Resolve named outputs and never rank the macro-label.

ID: GLB-022 · Profile: Pashto, Pakistan and Afghanistan profiles

Tag: ps-Arab-PK; ps-Arab-AF

Kind: cross_border_standard_profiles

Region: South/Central Asia

Band: >=50M combined mixed diagnostic

Disposition: unresolved_no_comparable_count

Population warning: Mixed denominator and dialect/orthography profiles cannot form one rank.

Next action: Resolve each profile; reuse a shared semantic source without blanket reach.

ID: GLB-023 · Profile: Sindhi, Pakistan

Tag: sd-Arab-PK

Kind: regional_standard

Region: South Asia

Band: 40-50M watch

Disposition: ranked_in_typed_view

Population warning: Shared RTL tooling is not intelligibility.

Next action: Audit school-to-tertiary and professional residual.

ID: GLB-024 · Profile: Latin American Spanish localized core

Tag: es-419 with country overlays

Kind: pluricentric_standard

Region: Americas

Band: >=250M

Disposition: supply_rich_negative_control

Population warning: Headline global totals are models and do not equal unique underserved readers.

Next action: Country-specific grades 3-9 recovery and exact professional/frontier gaps.

ID: GLB-025 · Profile: Brazilian Portuguese

Tag: pt-BR

Kind: national_pluricentric_standard

Region: Americas

Band: >=200M territory proxy

Disposition: supply_rich_negative_control

Population warning: Brazil population is not a Portuguese-reader count; global Portuguese totals lack disclosed proficiency method.

Next action: Build lawful offline teacher-independent bundles and exact research residuals.

ID: GLB-026 · Profile: Global/French regional localized standards

Tag: fr-Latn with country overlays

Kind: learned_lingua_franca

Region: Global

Band: >=250M institutional functional model

Disposition: supply_rich_negative_control

Population warning: Modeled Francophone totals include heterogeneous competence and schooling.

Next action: Audit country/stage-specific residuals and local-language bridges.

ID: GLB-027 · Profile: Russian learned standard

Tag: ru-Cyrl-RU; localized successor-state overlays

Kind: learned_lingua_franca

Region: Europe/Central Asia

Band: >=100M functional

Disposition: supply_rich_negative_control

Population warning: Knowledge may mean speaking only; no automatic written competence or ex-Soviet community coverage.

Next action: Open no-login professional/research-methods stack and exact frontier gaps.

ID: GLB-028 · Profile: German, Germany

Tag: de-Latn-DE

Kind: national_standard

Region: Europe

Band: >=50M home-language

Disposition: supply_rich_negative_control

Population warning: Home use is not a literacy test and does not cover all Germanic minorities.

Next action: Only exact open/accessibility/frontier residuals; do not use for Germanic minority coverage.

ID: GLB-029 · Profile: Italian, Italy

Tag: it-Latn-IT

Kind: national_standard

Region: Europe

Band: >=50M derived L1

Disposition: supply_rich_negative_control

Population warning: No direct Italian literacy census in located source.

Next action: Audit only exact open/accessibility/professional/frontier residuals.

ID: GLB-030 · Profile: Polish

Tag: pl-Latn-PL

Kind: national_standard

Region: Europe

Band: 40-50M watch

Disposition: supply_rich_negative_control

Population warning: Rounded survey shares are not exact population cells.

Next action: Exact residual audit only.

ID: GLB-031 · Profile: Modern Standard Arabic plus locale scaffolds

Tag: arb-Arab; country spoken-language layers

Kind: learned_written_standard

Region: MENA/Global

Band: >=250M umbrella

Disposition: forward_residual_audit

Population warning: Nobody's ordinary home language can be presumed to be MSA; Arabic umbrella varieties are not one comfort population.

Next action: Country-specific MSA-literacy plus local-spoken scaffolds, emergency and research pathways.

ID: GLB-032 · Profile: Egyptian Arabic scaffold

Tag: arz-Arab-EG; Saidi and other profiles separate

Kind: spoken_scaffold

Region: North Africa

Band: >=50M recall

Disposition: shared_core_only

Population warning: Egypt territory is not Egyptian-Arabic reader population; regional varieties remain.

Next action: Specify oral/audio/plain-language functions and exact orthography before reach credit.

ID: GLB-033 · Profile: Levantine Arabic profiles

Tag: apc-Arab with SY/LB/JO/PS locale layers

Kind: spoken_scaffold

Region: West Asia

Band: 40-50M watch

Disposition: shared_core_only

Population warning: Conflict/displacement and country varieties prevent one additive comfort count.

Next action: Build country-specific emergency and spoken-to-MSA bridges.

ID: GLB-034 · Profile: Iranian Persian

Tag: fa-Arab-IR

Kind: national_standard

Region: West/Central Asia

Band: >=50M recall

Disposition: ranked_in_typed_view

Population warning: Do not merge Dari and Tajik populations or scripts.

Next action: Resolve denominator and exact tertiary/professional/frontier residual.

ID: GLB-035 · Profile: Turkish, Turkey

Tag: tr-Latn-TR

Kind: national_standard

Region: West Asia/Europe

Band: >=50M territory/official proxy

Disposition: ranked_in_typed_view

Population warning: Territory and language-neutral literacy are not Turkish reader counts.

Next action: Rank only in territory/proxy view; audit exact open/accessibility/frontier residual.

ID: GLB-036 · Profile: Standard Kiswahili with country layers

Tag: swh-Latn-TZ; swh-Latn-KE; swh-Latn-UG; swc-Latn-CD

Kind: learned_lingua_franca

Region: Eastern/Central Africa

Band: >=100M gross regional functional estimate

Disposition: ranked_in_typed_view

Population warning: Global headline is not a deduplicated written-reader count; DRC Congo Swahili is distinct.

Next action: DRC early-grade/local variety and Uganda L2 first; later bilingual STEM/TVET/research.

ID: GLB-037 · Profile: Hausa Boko country outputs

Tag: ha-Latn-NG; ha-Latn-NE

Kind: cross_border_standard

Region: West Africa

Band: 40-50M watch L1; >=50M gross functional estimates

Disposition: unresolved_no_comparable_count

Population warning: 60M and 94M source universes differ; optional Ajami has separate unknown readership.

Next action: Resolve denominator; foundational/functional boko, country bridges, optional Ajami/audio.

ID: GLB-038 · Profile: Nigerian Pidgin / Naijá

Tag: pcm-Latn-NG

Kind: oral_lingua_franca

Region: West Africa

Band: 40-50M watch; >50M oral estimate

Disposition: unresolved_no_comparable_count

Population warning: Nigeria territory and oral users do not establish standardized written readership.

Next action: Audio/basic-phone practical modules first; measure orthography uptake before text rank.

ID: GLB-039 · Profile: Bahasa Indonesia

Tag: id-Latn-ID

Kind: national_standard

Region: Southeast Asia

Band: >=200M functional

Disposition: forward_residual_audit

Population warning: Oral ability is not written academic comfort; overlaps Javanese and other home languages.

Next action: No duplicate logic/basic math; finish exact program roles then evidence-to-practice/research core.

ID: GLB-040 · Profile: Javanese

Tag: jv-Latn-ID

Kind: regional_language

Region: Southeast Asia

Band: >=50M home-language

Disposition: ranked_in_typed_view

Population warning: Explicit subset/overlap edge with Indonesian is mandatory.

Next action: Bilingual/audio terminology transfer before duplicating the full advanced corpus.

ID: GLB-041 · Profile: Vietnamese

Tag: vi-Latn-VN

Kind: national_standard

Region: Southeast Asia

Band: >=50M speaker/territory

Disposition: ranked_in_typed_view

Population warning: Territory and language reach are distinct; minority-language early grades remain separate.

Next action: Coherent TVET-university data, green industry, health, MSME and research-method packages.

ID: GLB-042 · Profile: Thai

Tag: th-Thai-TH

Kind: national_standard

Region: Southeast Asia

Band: >=50M home/functional

Disposition: ranked_in_typed_view

Population warning: Derived union assumptions and minority-language overlap must remain visible.

Next action: Audit exact secondary/TVET/tertiary/open/accessibility gaps.

ID: GLB-043 · Profile: Japanese, Japan

Tag: ja-Jpan-JP

Kind: national_standard

Region: Eastern Asia

Band: >=100M territory ceiling

Disposition: forward_residual_audit

Population warning: Territory is not Japanese-language or underserved-reader count.

Next action: Accessible grades 5-9 bridge, research methods, then exact advanced algebraic-geometry gaps.

ID: GLB-044 · Profile: Korean, Republic of Korea

Tag: ko-Kore-KR

Kind: national_standard

Region: Eastern Asia

Band: >=50M territory ceiling

Disposition: forward_residual_audit

Population warning: Territory is not a Korean reader count and excludes DPRK conditions.

Next action: Offline life-capability and research reproducibility packages; exact advanced residual audit.

ID: GLB-045 · Profile: Korean, DPRK

Tag: ko-Kore-KP

Kind: national_standard_locale

Region: Eastern Asia

Band: regional constituent

Disposition: unresolved_no_comparable_count

Population warning: Do not infer DPRK access from ROK platforms or combine populations silently.

Next action: Obtain authoritative current supply/connectivity/access evidence before ranking.

ID: GLB-046 · Profile: Filipino, Philippines

Tag: fil-Latn-PH

Kind: learned_national_standard

Region: Southeast Asia

Band: >=50M functional recall; >100M territory ceiling

Disposition: unresolved_no_comparable_count

Population warning: Filipino is not every learner's L1 and Tagalog household counts are not national competence.

Next action: Resolve functional denominator; Filipino-English STEM-Kabuhayan with replaceable home-language tracks.

ID: GLB-047 · Profile: Burmese

Tag: my-Mymr-MM

Kind: national_standard

Region: Southeast Asia

Band: >=50M territory; functional below threshold

Disposition: ranked_in_typed_view

Population warning: Territory includes non-Burmese L1 communities.

Next action: Emergency/offline foundational and transition packages with exact minority-language overlays.

ID: GLB-048 · Profile: Amharic

Tag: am-Ethi-ET

Kind: national_federal_working_standard

Region: Horn of Africa

Band: >=50M broad school/speaker estimate; L1 below

Disposition: ranked_in_typed_view

Population warning: Broad estimate is not an exact written-reader count.

Next action: Secondary-tertiary STEM, professional/research bridge and offline delivery.

ID: GLB-049 · Profile: Oromo exact variety profiles

Tag: gaz/hae/gax/orc; locale outputs unresolved

Kind: macrolanguage_resolution

Region: Horn of Africa

Band: large regional; exact L1 below 50M in old census

Disposition: unresolved_no_comparable_count

Population warning: Oromia territory and Oromo macrolanguage are not one edition population.

Next action: Resolve exact profiles and secondary/agriculture/health need.

ID: GLB-050 · Profile: Standard Yoruba

Tag: yo-Latn-NG

Kind: regional_standard

Region: West Africa

Band: near/above 50M range

Disposition: unresolved_no_comparable_count

Population warning: Nigeria territory and unjoined literacy percentages cannot be multiplied without matching strata.

Next action: Resolve population/literacy join; STEM, health, finance and agriculture sequence.

ID: GLB-051 · Profile: Standard Igbo

Tag: ig-Latn-NG

Kind: regional_standard

Region: West Africa

Band: material 25-33M

Disposition: unresolved_no_comparable_count

Population warning: Continuum and standardized literacy require evidence.

Next action: Resolve standard/profile and exact foundational-to-tertiary residual.

ID: GLB-052 · Profile: Bhojpuri, India

Tag: bho-Deva-IN

Kind: regional_language_variety_set

Region: South Asia

Band: >=50M exact named L1 return

Disposition: ranked_in_typed_view

Population warning: The 50,579,447 count is a 2011 India mother-tongue return nested within the 528,347,193 Census Hindi umbrella; it is not a current, all-variety, literacy, academic-comfort, India-plus-Nepal, or diaspora count. Devanagari is a production default, not proof of one transregional standard.

Next action: Declare a versioned Devanagari editorial register, inventory exact current supply, and commission the missing teacher-independent phone/offline foundational-numeracy-through-algebra sequence with Bhojpuri-to-Hindi bridge terms; keep Nepal and Mauritius outputs separate.

ID: GLB-053 · Profile: Putonghua, Mainland spoken standard

Tag: cmn-Hans-CN for aligned text/audio; cmn-CN for speech-only metadata

Kind: learned_spoken_standard

Region: Eastern Asia

Band: >=1B derived functional sensitivity

Disposition: forward_residual_audit

Population warning: The 1,139,587,786 sensitivity is 80.72% of the mainland census denominator, not an official speaker count, L1 count, home-language count, literacy measure or academic-comfort count. It is wholly nonadditive with zh-Hans-CN and all Sinitic profiles.

Next action: Register spoken/audio outputs separately, preserve text/audio alignment, and award no direct written-reader or Sinitic-variety reach from this profile.

ID: GLB-054 · Profile: Malaysian Malay

Tag: ms-Latn-MY

Kind: national_pluricentric_standard

Region: Southeast Asia

Band: regional constituent of >=250M architecture

Disposition: forward_residual_audit

Population warning: The rounded 32.4M Malaysia territory population and 20.6M Bumiputera count are not Malay-reader counts. Indonesian contributes zero direct Malaysian demographic reach without directed evidence.

Next action: Reuse the completed Indonesian semantic and terminology assets, but produce and audit an explicit ms-Latn-MY locale output; keep Brunei Rumi/Jawi derivatives separate.

Appendix E. Bridge and shared-production architectures

ID: ARCH-001 · Architecture: Standard Written Chinese localized package

Mechanism: learned_written_standard

Named outputs: zh-Hans-CN; zh-Hant-TW; zh-Hant-HK; zh-Hant-MO; zh-Hans-SG; optional zh-Hans-MY

Diagnostic universe: Mainland territory >1.4B plus separate locale cells

Direct reach status: No blanket spoken-Sinitic reach; named written-standard outputs only

Compute reuse: Very high semantic/toolchain reuse

Exclusions: Wu, Yue vernacular, Taigi, Hakka and other Sinitic products remain separate

Next measurement: Measure locale-specific source/QA token reuse and written academic comfort

ID: ARCH-002 · Architecture: South Asian multilingual production system

Mechanism: parallel_semantic_core_compute_only

Named outputs: separate India/Pakistan/Bangladesh language-script outputs

Diagnostic universe: >1.35B current disjoint planning envelope

Direct reach status: Zero direct reader reach of the core

Compute reuse: Potentially very high shared formula/semantic/accessibility QA reuse

Exclusions: No pan-South-Asian reader surface or genealogy multiplier

Next measurement: Measure core, locale, script and QA tokens per named output

ID: ARCH-003 · Architecture: Global English overlap/accessibility baseline

Mechanism: learned_lingua_franca

Named outputs: plain English; locale overlays; bilingual glossaries

Diagnostic universe: >2B institutional first/additional-language diagnostic

Direct reach status: Unmeasured academic comfort; never unique reach by default

Compute reuse: High source reuse and accessibility leverage

Exclusions: Does not replace local-language editions

Next measurement: Measure stage/subject comfort and exact plain/offline access gain

ID: ARCH-004 · Architecture: Arabic MSA plus spoken-language scaffolds

Mechanism: learned_written_standard

Named outputs: arb written core plus Egyptian/Levantine/Iraqi/Gulf/Maghrebi/Yemeni/Sudanese/Hassaniya layers

Diagnostic universe: 400-450M umbrella diagnostic

Direct reach status: Named outputs only; MSA not L1

Compute reuse: High formal-core and RTL reuse

Exclusions: Non-Arabic L1 communities and dialect comfort remain separate

Next measurement: Measure local oral scaffold effect by stage/task

ID: ARCH-005 · Architecture: Romance localized/intercomprehension core

Mechanism: natural_intercomprehension_and_pluricentric_reuse

Named outputs: Spanish; Portuguese; French; optional research bridge

Diagnostic universe: Nonadditive >1B headline speaker estimates

Direct reach status: Only directed written comprehension receives reach

Compute reuse: High terminology/source reuse

Exclusions: No one pan-Romance replacement or summed audience

Next measurement: Measure mathematical expert/novice written comprehension and locale token reuse

ID: ARCH-006 · Architecture: Hindi-Urdu two-script core

Mechanism: pluricentric_shared_core

Named outputs: hi-Deva-IN; ur-Aran-IN; ur-Aran-PK

Diagnostic universe: 395,205,166 exact named L1 cells

Direct reach status: No demonstrated untrained academic-text cross-script reach

Compute reuse: High semantic/formula reuse

Exclusions: Hindi umbrella subsidiaries remain separate

Next measurement: Measure transliteration/locale QA and directed educational comprehension

ID: ARCH-007 · Architecture: Indonesian-Malaysian Malay package

Mechanism: pluricentric_shared_core

Named outputs: id-Latn-ID; ms-Latn-MY; Brunei/Jawi named derivatives

Diagnostic universe: ~274M functional diagnostic; ~319M territory ceiling

Direct reach status: Only each named locale population receives direct reach

Compute reuse: Very high MABBIM/terminology reuse

Exclusions: Javanese, Sundanese, Madurese and other languages excluded

Next measurement: Measure Malaysian/Bruneian locale delta after completed Indonesian source

ID: ARCH-008 · Architecture: Bangla Bangladesh-India shared core

Mechanism: pluricentric_shared_core

Named outputs: bn-Beng-BD; bn-Beng-IN

Diagnostic universe: ~264.6M L1 diagnostic

Direct reach status: Two distinct locale outputs

Compute reuse: High same-script semantic reuse

Exclusions: No locale-neutral curriculum or professional overlay

Next measurement: Measure source reuse and locale-specific terminology/examples

ID: ARCH-009 · Architecture: Russian learned-standard overlap

Mechanism: learned_lingua_franca

Named outputs: ru-Cyrl locale/localization outputs

Diagnostic universe: >250M institutional diagnostic

Direct reach status: Unique residual unmeasured

Compute reuse: High source/localization reuse

Exclusions: No blanket ex-Soviet/Indigenous/displaced coverage

Next measurement: Measure academic comfort and local-language nonoverlap

ID: ARCH-010 · Architecture: Standard Kiswahili country package

Mechanism: learned_lingua_franca

Named outputs: swh-TZ; swh-KE; swh-UG; swc-CD regional outputs

Diagnostic universe: >200M institutional diagnostic

Direct reach status: Named country/variety outputs only

Compute reuse: High standard-core reuse

Exclusions: DRC varieties and non-Swahili L1 communities remain

Next measurement: Measure country-layer delta and written comfort

ID: ARCH-011 · Architecture: Dravidian parallel pipeline

Mechanism: parallel_semantic_core_compute_only

Named outputs: te; ta-IN/ta-LK; kn; ml outputs

Diagnostic universe: 228,084,103 exact Indian L1 components before Sri Lanka

Direct reach status: Zero core reach

Compute reuse: Shared formula, layout and QA only unless measured

Exclusions: Genealogy supplies no comprehension multiplier

Next measurement: Measure actual reused tokens across named scripts/outputs

ID: ARCH-012 · Architecture: Slavic/Interslavic supplementary bridge

Mechanism: constructed_bridge

Named outputs: dual-script Interslavic plus native outputs

Diagnostic universe: ~266-320M family ceiling

Direct reach status: Current short-task evidence is not demographic reach

Compute reuse: Terminology/semantic reuse plausible

Exclusions: Russian/Ukrainian bias and communities not tested remain explicit

Next measurement: Test mathematics by expertise, script and task; measure production reuse

ID: ARCH-013 · Architecture: Turkic branch production

Mechanism: parallel_semantic_core_and_constructed_hypothesis

Named outputs: Oghuz, Kipchak, Karluk named outputs

Diagnostic universe: ~200M family; Oghuz ~110-120M

Direct reach status: No pan-Turkic direct reach

Compute reuse: Branch terminology/model reuse plausible

Exclusions: No family-wide standard or comprehension

Next measurement: Measure per-branch reuse and directed text comprehension

ID: ARCH-014 · Architecture: Persian-Dari-Tajik core

Mechanism: pluricentric_shared_core

Named outputs: fa-IR; prs-AF; tg-Cyrl-TJ

Diagnostic universe: ~102M functional diagnostic; ~147M territory ceiling

Direct reach status: Named outputs only

Compute reuse: High semantic reuse with script/locale deltas

Exclusions: No one script/register or additive comfort count

Next measurement: Resolve exact counts and measure script/terminology delta

ID: ARCH-015 · Architecture: Punjabi dual-script core

Mechanism: dual_script_register

Named outputs: pa-Guru-IN; pa-Arab-PK

Diagnostic universe: 120,059,639 exact source-labelled L1 cells

Direct reach status: Separate scripts/locale reach only

Compute reuse: High semantic and transliteration assistance

Exclusions: Pakistan variety identity and cross-script literacy unresolved

Next measurement: Resolve Pakistan profile and measure post-edit/QA cost

ID: ARCH-016 · Architecture: Hausa country/orthography package

Mechanism: learned_lingua_franca_shared_core

Named outputs: ha-Latn-NG; ha-Latn-NE; optional established Ajami layers

Diagnostic universe: Cross-border >=100M plausible but unverified

Direct reach status: No bankable cross-border written reach

Compute reuse: Strong shared-source potential

Exclusions: No universal Ajami or territory-wide reach

Next measurement: Obtain compatible population/literacy evidence and measure country delta

Appendix F. Reproducibility identities

Artifact	Rows	Bytes	SHA-256
TOP_100.csv	100	406614	C5ACDEE9A09805A4B0C22346655C429FE7906E927512A8432F74DCE3183CBB34
TOP_10.csv	10	46175	616E2571B02A3F1B29012479E01405B4E2169541DB4AD390CB3BB603D34B5103
top100_needs_assignment_v2.csv	100	280635	E3098F019722839E0CB64FBE4740E426F907A5FDD568A3E61E715F23D881CBA2
top100_interlanguage_overlap_crosswalk.csv	100	112187	4C5021F7E5E6CBF642574A2AAA0BB2E3CD70CA408129E888309F5DF13C0D4E79
expected_language_profiles_v1_1.csv	54	28762	103AC91B4BC485EAC1849A0F4ADC4D398869D5A32AC697BBE9E46875D131E5D2
bridge_shared_core_hypotheses_v1_1.csv	16	6976	34B85DEB823977966C403575AA3119B5588644156373177E6645AE3EAD0EB7BB
profile_population_estimates_supplement_v1_1.csv	38	33515	0BA2DDC28306F9A79A256F47F80F0D2AEBB99EB34E80C782B731E1E316E9852B
expected_profile_population_joins_v1_1.csv	91	100823	D949CDA66DCCD469DC3DB706933412B6C93C043ED83C48C32673C35385C44548
high_reach_education_needs_v1_1.csv	20	33593	A41FBAC87B2DF2E3876392E7AA199DB0DAE7803FE1629D6B61CBD229343391B6
table_c_forward_residual_commissioning.csv	12	13418	0755B3AE7ED06EFC5D7505B160C1E7F5F8825DEBC04569E8D925443E9393273C
V1_1_DATA_PIPELINE_REBUILD_20260831.json	—	26311	B59154152CC35DF16B7F26D0233C4FB4129F5AB99E121676CA0CF6DCE7A546F7
top100_companions_v1_1_validation.json	—	1550	605A3E7E8D3E516D811FC20902D787143A19E20CD73C17388403C3D7C949E638
allocation_policy_v1_1_validation.json	—	4473023	DA7831A1C4F01D3FC9D650C4C0F820E8B38BA2412A1FEC34D05C966BD7DC1B1C

Artifact	Rows	Bytes	SHA-256
opportunity_planning_sensitivity_v1_1.csv	792	127265	B3583CB76859E9B62D6646CE6349F78FC815634DA67230468E3B09E61D394589
asia_core.json	—	89458	8068948F3296B2EDAE599C4126C07F6F557C1958799401EC4F0E0D2CFA7D8F30
large_profiles.json	—	84832	28EACA5383EB4011A0A8BAA3F85C9513D2842FFBB27A03C4F080D714B250874D
component_work_ownership_v1_1.json	—	12636	8917A77C461A3196CE2B1DF2F9A256C36DB3C358EFD8B5727F486D2F4B7D025F

The source-to-claim research memos are `JAPAN_EQUAL_BASIS_RESEARCH_20260831.md` (SHA-256 `AE40A7DB56DB5F66A6F9DF6ADFFA21B31B7DF1629A012EBAB4CEC8651DC7320C`), `SIMPLIFIED_CHINESE_RESIDUAL_RESEARCH_20260831.md` (SHA-256 `AA62604919E63BEB0DA8DA3CEA39AAD31B884450E82CA8D962806B8B8932F109`), `GLOBAL_LARGE_PROFILE_AND_MODALITY_AUDIT_20260831.md` (SHA-256 `0D2C7583BA5A8F830C5959DDE4875DB3BE5DA920EA8F36CE5EFBB81D54B540B2`), and `INDONESIAN_PROGRAM_COMPUTE_AND_SCALE_AUDIT_20260831.md` (SHA-256 `4A93B4EDEC44A8D310B2DAC94B87EED60EA70096925068DB6A0A843569F56260`).

Appendix G. Scope-calibrated planning judgments

The following are literature-constrained commissioning judgments, not measured intervention effects or guaranteed positive floors. The population facts and curriculum/supply sources constrain the reasoning; the residual audience is an explicit inference. R already incorporates overlap, and no additional N multiplier is applied. Exact source chains, confidence, assumptions and threshold tests are preserved in `asia_core.json` and `large_profiles.json`.

Action: NAT-121

Position: position 32

R: R3 (range 2–3)

S: S2 (range 2–3)

A: A4 (range 3–4)

N: N2 (range 2–3)

Exact planning scope: The selected six-course Evidence-to-Practice and Research Core: research design/open science; applied statistics/causal reasoning in R/Python; reproducible data/AI/cybersecurity; evidence appraisal; public health/One Health; climate-smart agriculture/disaster risk. The same person taking several courses counts once.

Reader-band reasoning: A cross-disciplinary continuation serves a standing tertiary/professional cohort, not all Indonesian speakers. Official higher-education statistics place the enrolled cohort near ten million; the established mathematics program supplies prerequisites and reusable delivery infrastructure. WHO's separately reported 574,343 nurses supports a concrete professional route but is not added to enrollment. Hundreds of thousands to low millions gaining materially easier independent access is a plausible planning envelope; no source measures that residual directly. R4 would require over ten million newly served, approximately the entire enrolled cohort before any overlap discount, so it is not assigned to this six-course scope.

Action: NAT-122

Position: position 34

R: R3 (range 2–3)

S: S2 (range 1–2)

A: A4 (range 3–4)

N: N2 (range 1–2)

Exact planning scope: Complete Calculus I CALC1-0031 through CALC1-0055, continue Calculus II/III and Statistics, and add semantic/offline/accessible derivatives to completed corpora. The later school, 144-hour vocational, adult-finance and frontier proposals do not contribute extra audience to this score.

Reader-band reasoning: The 48.7257m higher-education enrollment provides a relevant gross scale, but substantial native-language calculus/statistics supply and completed Open Logic/algebra imply a much smaller residual. A multi-title no-login, downloadable, exercise-complete and semantic-accessibility package can plausibly improve access for hundreds of thousands to low millions of prepared students and independent learners; this remains a planning judgment. Over ten million would require demonstrating a much broader distinct residual, which the current exact package evidence does not establish.

Action: NAT-123

Position: position 37

R: R2 (range 1–3)

S: S2 (range 2–3)

A: A4 (range 3–4)

N: N2 (range 2–3)

Exact planning scope: The selected Japanese Open Mathematics Access and Research Bridge: accessible self-study diagnostics/prerequisite repair, full solutions, easy-Japanese/furigana and multilingual terminology for non-attending pupils, adult re-entry and pupils needing Japanese-language instruction. Later title-specific EGA/SGA/FGA translations are not counted as a mass audience.

Reader-band reasoning: Officially identified cohorts include 353,970 non-attending compulsory pupils, 84,759 needing Japanese instruction and an approximately 7.47m low-numeracy adult exposure. These overlap and neither outcome directly measures language-caused deprivation. Tens of thousands to low millions benefiting from the specific no-login accessible/re-entry route is plausible; the central planning band is hundreds of thousands because school-textbook supply is strong and most adults require more than language substitution alone. Adult re-entry supports an R3 favorable scenario, not an R4 or language-wide negative control.

Action: SHC-BN

Position: position 1

R: R3 (range 2–3)

S: S3 (range 2–3)

A: A3 (range 2–3)

N: N3 (range 2–4)

Exact planning scope: Residual SHC-BN only: Bangladesh 24-week caregiver-mediated early-childhood numeracy/school-readiness for children 36-59 months, plus Indian Bengali secondary/TVET workshop mathematics, safety, coding, finance and entrepreneurship. Link NAT-001 for grades 2-5 recovery; link NAT-002 for Indian grades 3-8 catch-up rather than duplicate either spine.

Reader-band reasoning: Base R3 is a joint residual-package judgment: the Indian Bengali TVET component alone has a low-million central opportunity, with a smaller additional Bangladesh preschool component. It is not R3 because two labels were summed. Low R2 reflects weaker residual use in both settings; favorable R3 reflects several-million joint use, not the Cartesian maximum of both broad component bands. R4 is not proved impossible but is no longer a retained favorable scenario for this narrower first output. No primary recovery audience is included.

Action: IL-HU

Position: position 57

R: R2 (range 2–3)

S: S3 (range 2–3)

A: A3 (range 2–3)

N: N3 (range 2–4)

Exact planning scope: Residual IL-HU only: two separately localized Urdu first-year disciplinary and research-methods outputs, ur-Aran-IN and ur-Aran-PK. Reuse the separately owned NAT-124 Hindi semantic/formula source; do not commission or count a new Hindi output.

Reader-band reasoning: Base R2 is a direct joint judgment of hundreds-of-thousands newly comfortable Urdu users across the two localized first-year/methods packages, not an arithmetic consequence of two R2 labels. Each component could occupy different parts of its broad band; together they may cross 1m, so favorable R3 is explicit. The cautious joint endpoint R2 is consistent with Pakistan's own cautious R2 component. Hindi's much larger population and professional reach are excluded completely from this residual score.

Action: NAT-125

Position: position 4

R: R3 (range 2–4)

S: S3 (range 3–4)

A: A3 (range 2–3)

N: N3 (range 2–4)

Exact planning scope: Declared versioned Devanagari Bhojpuri editorial register with teacher-independent foundational numeracy through algebra, full worked answers, offline/phone/audio delivery and explicit Bhojpuri-to-Hindi academic terminology transfer. Bihar/eastern-UP overlays; no Nepal or diaspora audience.

Reader-band reasoning: The 50.579m exact mother-tongue return and absence of an evidenced continuous Bhojpuri mathematics/STEM sequence support a substantial plausible first-language scaffold opportunity. Hindi-medium schooling does not establish comfortable academic comprehension, but nor can its overlap be treated as zero. Audio and explicit transfer reduce the written-literacy barrier. A low-million standing opportunity is a defensible provisional base; the wide range reflects unmeasured Hindi comfort, age/stage match and oral-to-written uptake.

Action: NAT-001

Position: position 51

R: R3 (range 2–3)

S: S3 (range 2–3)

A: A3 (range 2–3)

N: N3 (range 2–4)

Exact planning scope: Bangladesh Bangla grades 2-5 number sense, operations, place value, fractions, measurement and patterns with diagnostics and full worked answers, including only caregiver instructions supporting these same grades. Excludes preschool 36-59 months and Indian Bengali outputs.

Reader-band reasoning: Base R3 is a low-million planning opportunity for a four-grade recovery kit within the large Bangladesh child cohort, supported by directly observed foundational deficits and missing practice/feedback continuity. The former Bangladesh component included preschool/ECE; that extra audience is now excluded. R4 is not retained for this narrower initial kit, and no grade-population interpolation or arbitrary percentage is used.

Action: NAT-124

Position: position 7

R: R3 (range 2–4)

S: S3 (range 2–3)

A: A4 (range 3–4)

N: N3 (range 2–3)

Exact planning scope: Standard Hindi hi-Deva-IN first-year physics, chemistry, biology, statistics, computing, economics and management, followed by research design, R/Python and open-science methods. Exact HRN-004; no Urdu/Bhojpuri or already completed formal-reasoning audience.

Reader-band reasoning: Base R3 is a low-million Standard Hindi professional/tertiary continuity opportunity, not the whole 322.230m mother-tongue cell. Extensive school supply and completed formal reasoning are credited. R4 is favorable for the same selected multidisciplinary and methods route including independent re-entry/professional users, not for an unscoped all-education portfolio; no numerical fraction is chosen.

Action: SHC-HAUSA

Position: position 5

R: R3 (range 2–4)

S: S3 (range 2–3)

A: A2 (range 2–3)

N: N3 (range 3–4)

Exact planning scope: Current first package only: Nigeria/Niger boko foundational literacy and numeracy, with separately localized English/French bridges and print/audio/IVR; optional Ajami is a separate named layer. Potential learners after the package's own entry bridge, not already fluent readers and not future TVET users.

Reader-band reasoning: A tens-of-millions Hausa speech community and independently evidenced foundational-learning crisis make a million-scale joint cohort plausible for a full foundational package. R3 is the base judgment; R4 is a favorable extension to adult beginning readers and both country outputs, not a finding that 10–50M currently read boko. No R5 credit.

Action: SHC-SWAHILI

Position: position 19

R: R2 (range 2–3)

S: S2 (range 1–3)

A: A2 (range 2–3)

N: N2 (range 2–3)

Exact planning scope: Current first package only: DRC grades 1–3 Kiswahili completion/adaptation with French bridge and regional-variety overlays; Uganda P4 Kiswahili-L2 support with offline audio. Tanzania/Kenya secondary or advanced editions are follow-on, not counted here.

Reader-band reasoning: The current package targets only two grade/country routes, so whole-family 200M speech estimates and legacy Tanzanian R4 are irrelevant to its first-output audience. A hundred-thousand-to-million-scale opportunity is plausible across these large systems but an exact language-by-grade-by-supply joint denominator is absent.

Action: IL-AR

Position: position 15

R: R3 (range 2–4)

S: S2 (range 2–3)

A: A2 (range 2–3)

N: N3 (range 3–4)

Exact planning scope: Current first package only: vowelled MSA foundational sequence with named country spoken narration, plus country-specific crisis catch-up literacy/science/health/WASH. Count only children whose spoken layer and entry route are actually supplied; later MSA professional/research editions are excluded.

Reader-band reasoning: Regional crisis and foundational-learning cohorts are large enough to support a provisional million-scale planning opportunity, while the mechanism is exact spoken-to-MSA scaffolding, not generic MSA availability. R4 is a favorable scenario across multiple completed country outputs, not a 400M blanket-reader assumption.

Action: NAT-028

Position: position 14

R: R3 (range 2–3)

S: S2 (range 2–3)

A: A4 (range 3–4)

N: N3 (range 2–3)

Exact planning scope: Current five Vietnamese applied TVET/university courses in data/digital work, green agriculture/aquaculture, health evidence, MSME finance and applied biology/chemistry; their worked self-study material, not later postgraduate/frontier courses.

Reader-band reasoning: Five practically broad courses can plausibly serve a million-scale residual from large employed and education cohorts, but the older 89M language estimate is not their readership. The industrial/digital component alone has a 17.4M employed-sector context and documented skills need. R3 is a provisional joint-eligibility judgment, not an observed conversion rate.

Action: NAT-033

Position: position 52

R: R3 (range 2–3)

S: S2 (range 1–2)

A: A3 (range 3–4)

N: N2 (range 2–3)

Exact planning scope: Current 120 Filipino-English STEM-Kabuhayan micro-lessons for grades 7–12, ALS and TVET, with replaceable home-language tracks, worked answers and offline distribution. Not the whole Philippine population or later research spine.

Reader-band reasoning: Official secondary entry-grade registrations are already million-scale, and the package spans six grades plus ALS/TVET. Combined with offline-access gaps and the explicit bilingual design this supports a provisional R3, not proof all secondary learners lack resources.

Action: NAT-003

Position: position 2

R: R3 (range 2–3)

S: S3 (range 2–3)

A: A4 (range 3–4)

N: N3 (range 3–4)

Exact planning scope: Current applied secondary/TVET/first-year STEM, coding, financial safety and entrepreneurship package with Telugu-English terminology, Andhra Pradesh/Telangana crosswalks, worked solutions and offline simulations.

Reader-band reasoning: The aged census language category minus reported English-or-Hindi subsidiary-language union establishes a large no-reported-bridge context, not an academically ready or newly served count. Combining that with the exact stage discontinuity supports a 100k–10M planning interval, with the stated baseline; legacy whole-OpenLogic R4 is not copied onto this first package.

Action: NAT-006

Position: position 35

R: R3 (range 2–3)

S: S2 (range 2–3)

A: A4 (range 3–4)

N: N2 (range 2–3)

Exact planning scope: Current applied science/workshop/computing/finance/entrepreneurship package for secondary/TVET/first-year transition, not later statewide agriculture or research courses.

Reader-band reasoning: The aged census language category minus reported English-or-Hindi subsidiary-language union establishes a large no-reported-bridge context, not an academically ready or newly served count. Combining that with the exact stage discontinuity supports a 100k–10M planning interval, with the stated baseline; legacy whole-OpenLogic R4 is not copied onto this first package.

Action: NAT-004

Position: position 21

R: R2 (range 2–3)

S: S2 (range 2–3)

A: A4 (range 3–4)

N: N3 (range 3–4)

Exact planning scope: Current Tamil-English first-year STEM/computing/management bridge with worked problems, offline simulations and introductory research methods. Later professional education and all school grades are outside this R.

Reader-band reasoning: The aged census language category minus reported English-or-Hindi subsidiary-language union establishes a large no-reported-bridge context, not an academically ready or newly served count. Combining that with the exact stage discontinuity supports a 100k–10M planning interval, with the stated baseline; legacy whole-OpenLogic R4 is not copied onto this first package.

Action: IL-PUNJABI

Position: outside the current Top 100; retained in the full candidate disposition

R: R2 (range 2–3)

S: S3 (range 2–3)

A: A3 (range 2–4)

N: N2 (range 2–3)

Exact planning scope: Current synchronized secondary/TVET/first-year STEM package with distinct Gurmukhi India and declared Pakistan Perso-Arabic outputs. Numerical interval below covers the currently resolved Indian component only; Pakistan's target-resolution ceiling is shown separately, not zeroed.

Reader-band reasoning: The resolved Indian component supports a provisional hundred-thousand-to-million-scale transition opportunity. Pakistan offers potentially much larger additional access, but its census Punjabi label does not resolve one written production variety; assigning the entire 88.9M to a particular ISO code would fabricate precision.

Action: GLB-044

Position: position 17

R: RU

S: S1 (range 1–2)

A: A4 (range 3–4)

N: N2 (range 1–2)

Exact planning scope: Current 36 offline life-capability modules in health, fraud/SME finance, agriculture/climate, practical science, computing and AI verification.

Reader-band reasoning: No source inspected here cross-tabulates the proposed residual package, language comfort and existing usable-resource access. An R2/R3 assignment based solely on a large industrialized population would be arbitrary. The profile remains a positive residual opportunity with RU, not R0 or a blanket negative control.

Action: GLB-024

Position: position 11

R: RU

S: S1 (range 1–2)

A: A4 (range 3–4)

N: N2 (range 1–2)

Exact planning scope: Current grades 3–9 mastery-recovery package with health/household finance/digital/climate applications and complete diagnostic feedback; country overlays required.

Reader-band reasoning: No source inspected here cross-tabulates the proposed residual package, language comfort and existing usable-resource access. An R2/R3 assignment based solely on a large industrialized population would be arbitrary. The profile remains a positive residual opportunity with RU, not R0 or a blanket negative control.

Action: GLB-025

Position: position 12

R: RU

S: S1 (range 1–2)

A: A4 (range 3–4)

N: N2 (range 1–2)

Exact planning scope: Current adult/vocational/professional self-study mastery/transition bundle built from reusable Brazilian sources; no-login/offline and semantic access.

Reader-band reasoning: No source inspected here cross-tabulates the proposed residual package, language comfort and existing usable-resource access. An R2/R3 assignment based solely on a large industrialized population would be arbitrary. The profile remains a positive residual opportunity with RU, not R0 or a blanket negative control.

Action: GLB-027

Position: position 9

R: RU

S: S1 (range 1–2)

A: A4 (range 3–4)

N: N2 (range 1–2)

Exact planning scope: Current statistics/causal-inference/reproducible Python-R/research-integrity/data-stewardship/AI-evidence course with applied tracks; later frontier texts excluded.

Reader-band reasoning: No source inspected here cross-tabulates the proposed residual package, language comfort and existing usable-resource access. An R2/R3 assignment based solely on a large industrialized population would be arbitrary. The profile remains a positive residual opportunity with RU, not R0 or a blanket negative control.

Sources: Several source checks materially changed package scope. The DRC ministry already lists Kiswahili early-grade pupil books, workbooks, teacher guides and literacy materials, so the proposal targets specified feedback, accessible/offline and regional-variety gaps rather than claiming all such textbooks absent (Ministry of National Education and New Citizenship, n.d.). UNICEF's 30-million figure concerns out-of-school children aged 5–18 across 12 MENA countries, not 30 million Arabic readers (UNICEF, 2025). Final BBS/UNICEF MICS 2025 reports foundational numeracy for 39.2% of ages 7–14, reading for 50.2%, and ECE attendance for 16.6% at ages 36–59 months; those outcome/attendance measures are not a causal count of language exclusion (Bangladesh Bureau of Statistics & UNICEF, n.d.).

Source: The Indonesian 2025 higher-education report extraction gives 10,754,154 enrolled students; the live portal shows a different snapshot near ten million. The 33.9-MB report was not page-audited in this bounded pass, so the exact extraction remains provisional and the snapshots are not averaged; only their order of magnitude informs the recommendation. Pakistan's indexed 2022-23 enrollment anchor is 1,964,147, not a measure of Urdu readership (Higher Education Commission, 2024, p. 202). Indian Hindi evidence is not independent Urdu evidence. These are source limits, not zero-access findings.